Unit 1

Measurement, Limits & Fits, Gauges

(Handwritten Notes)

Syllabus Topics Covered

1. Concept of measurement, feedback control, generalized measurement system
2. Units and standards
3. Measuring instrument characteristics: sensitivity, accuracy, precision, response
4. Errors, correction and calibration
5. Limits, fits and tolerances; hole & shaft basis; interchangeability
6. Limit gauges, Taylor's principle, gauge design
7. Linear measurement: vernier, micrometer, slip gauges, tool maker's microscope
8. Comparators: mechanical, pneumatic, electrical
9. Angular measurement: sine bar, bevel protractor, angle dekkor, rollers & spheres

1. Concept of Measurement

Metrology is the science of measurement. **Measurement** is the process of **comparing an unknown quantity with a known standard** of the same kind, to find how many times the standard is contained in the unknown quantity.

Result of a measurement

Measurement = Number x Unit (e.g., length = 25.4 x mm)

Objectives of metrology

- To evaluate **newly developed products** and judge their performance, and to determine the **process capabilities** .
- To determine the **measuring instrument capabilities** and ensure they are adequate for the application.

- To **minimize inspection cost** and the cost of rejection; to maintain **interchangeability** of parts; to **standardize** methods and the gauging practice.
- To obtain the **quality** level of the manufacturing process through **feedback** , and to establish the **traceability** of measurements to national standards.

Methods of measurement

Method	Description	Example
Direct	value obtained directly by comparing with a standard	steel rule, vernier caliper
Indirect	value computed from other measured quantities	area = l x b; sine bar angle
Comparative	compared with a known standard; only deviation shown	dial indicator + slip gauges
Absolute / fundamental	based on measurement of base quantities	dead-weight pressure gauge

Generalized measurement system

Any measuring system can be broken into **functional elements** (stages):

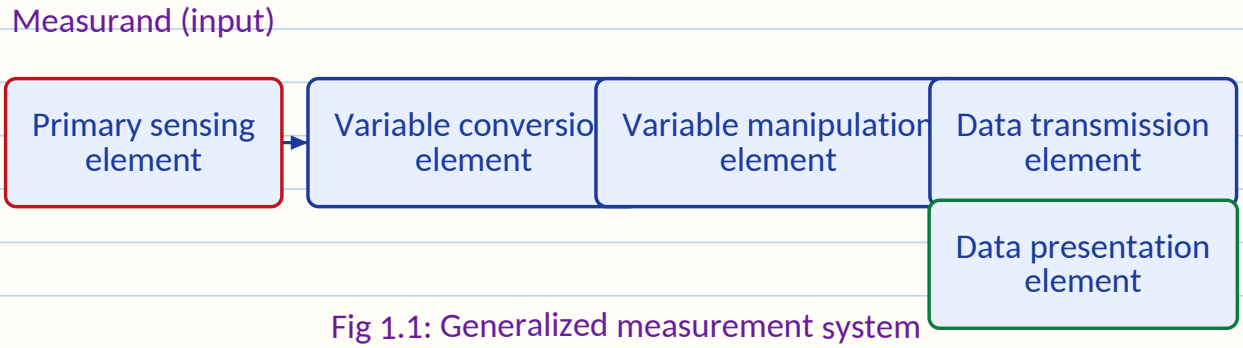


Fig 1.1: Generalized measurement system

1. **Primary sensing element** - senses the quantity (**measurand**) and gives output (e.g., bellows, bimetallic strip, Bourdon tube).
2. **Variable conversion element** - converts the output to a more convenient physical variable (e.g., strain gauge, LVDT).
3. **Variable manipulation element** - amplifies / modifies the signal (amplifier, gear train, lever).
4. **Data transmission element** - carries the signal from one place to another (cables, telemetry, linkages).
5. **Data presentation element** - displays the result (dial, scale, recorder, printer, CRO, digital display).

Concept of feedback control system

A **control system** keeps an output at a desired value by **feeding back** the measured output and comparing it with the reference (set point). Measurement is the **feedback element** of the loop.

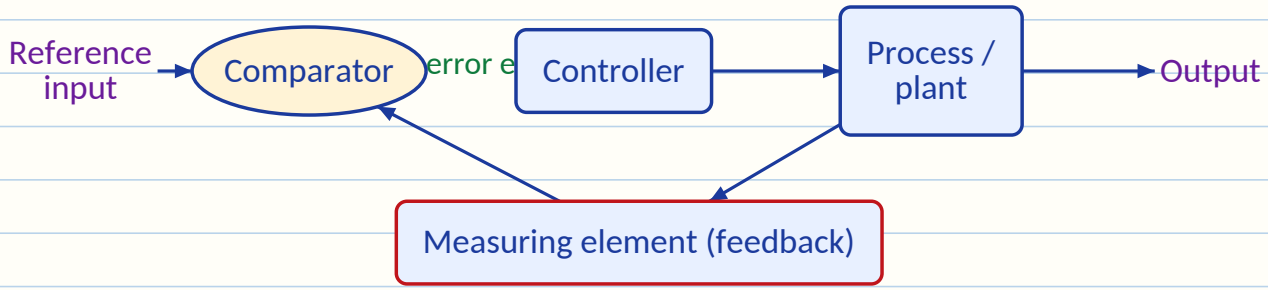


Fig 1.2: Feedback control system (closed loop)

- **Error = Reference - Feedback** . Controller acts to reduce the error. Examples: thermostat, speed governor, CNC machine position feedback by encoder.
- **Open-loop** system has no feedback (washing machine timer) - cannot correct disturbances.

2. Units and Standards

A **unit** is a definite magnitude of a physical quantity used for comparison. A **standard** is the physical embodiment of a unit. The **SI** (Systeme International) is the universally accepted system.

Base quantity	SI unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Luminous intensity	candela	cd
Amount of substance	mole	mol

- **Supplementary / derived** : plane angle - radian; solid angle - steradian; force - newton (N); pressure - pascal (Pa); energy - joule (J); power - watt (W).
- **Metre** (current definition): the length of the path travelled by light in vacuum during **1/299 792 458 s** . Earlier: international prototype (Pt-Ir bar, 1889) and

1,650,763.73 wavelengths of Kr-86 orange-red line (1960).

Types of standards

1. **Line standard** : length defined as the distance between **two engraved lines** on a bar (Imperial standard yard, International prototype metre).
2. **End standard** : length defined between **two flat parallel faces** (slip gauges, end bars, micrometer anvils).
3. **Wavelength standard** : length defined in terms of wavelength of light (krypton-86, laser) - **not** subject to wear or damage; reproducible anywhere.

Line standard	End standard
Distance between two lines	Distance between two flat faces
Quick and easy to use; parallax error	No parallax; wringing needed; better accuracy
Not so accurate (+-0.2 mm to 0.1 mm)	Highly accurate (to 0.0005 mm)
Subject to scale wear / marking errors	Faces wear with use; temperature effects
Steel rule, metre bar	Slip gauge, vernier jaws, micrometer

- **Primary** : kept at the national / international level, never used directly. **Secondary** : calibrated against primary, used to calibrate others. **Working** : used daily in workshops (slip gauges grade 2, gauges).

3. Measuring Instrument Characteristics

Term	Meaning
Sensitivity	ratio of change in output (pointer movement) to change in the measured input: S = dO/dI. Higher magnification -> higher sensitivity. Also the smallest input change that gives a perceptible output.
Readability	ease with which the readings can be taken, depends on the scale spacing, pointer thickness and parallax. Vernier / digital is easier to read.
Range	difference between the maximum and minimum values the instrument can measure (e.g., 0-25 mm micrometer). **Span** = max - min. **Least count (resolution)** = smallest reading possible.
Accuracy	closeness of the measured value to the **true value** . Expressed as +-% of full scale / reading.

Precision	closeness of **repeated readings** to each other - **repeatability** , not related to the true value.
Repeatability	ability of the instrument to give the same reading when the same quantity is measured repeatedly, under same conditions, short time interval.
Hysteresis	difference in reading for the same input when approached from above and below (friction, backlash).
Magnification	ratio of the length of the pointer (scale) movement to the change in size of the work. Linear: sensitivity = magnification.

Static and dynamic response

- **Static characteristics** : those for measurements of **constant or slowly varying** quantities - accuracy, sensitivity, hysteresis, range, resolution, linearity, repeatability.
- **Dynamic characteristics** : how the instrument responds to **rapidly changing** input:
  - **Speed of response** - how quickly the instrument reacts to a change in the measured quantity.
  - **Measuring lag** - delay in response (retardation type: begins at once; time-delay type: begins after a dead time).
  - **Fidelity** - ability to reproduce the input faithfully, without dynamic error.
  - **Dynamic error** - difference between the true value of the time-varying quantity and the value indicated (assuming zero static error).

4. Errors, Correction and Calibration

Error

Error = Measured value - True value . **Correction = -(Error)** , i.e. True value = Measured value + Correction.

- Absolute error = |measured - true|; Relative (percentage) error = (error / true value) x 100.

Types of errors

1. **Systematic (fixed) errors** : constant in magnitude and sign, **predictable** and can be **eliminated** by correction / calibration. Causes: faulty calibration, zero error, wear of instrument, temperature effect, **loading** errors, **Abbe's** error.

2. **Random errors** : vary in magnitude and sign **unpredictably** - caused by small changes in temperature, humidity, vibration, operator judgement. Cannot be eliminated; treated **statistically** (mean, standard deviation); reduced by averaging many readings.
3. **Gross errors** : human mistakes - wrong reading, wrong recording, misuse of instrument.

Systematic errors	Random errors
Same magnitude/sign each time	Vary in magnitude and sign
Can be determined and corrected	Cannot be corrected; estimated statistically
Affect accuracy	Affect precision
Eg. zero error, wrong calibration	Eg. thermal fluctuations, vibration

Calibration

Calibration is the process of comparing an instrument against a **standard of higher accuracy** under specified conditions, to determine its error and apply corrections or adjust it.

- Steps: select the standard -> apply known inputs over the range (up and down) -> record the instrument readings -> prepare the **calibration curve / table** -> compute errors, hysteresis -> adjust, certify and label the next calibration due date.
- **Traceability** : unbroken chain of calibrations to the national / international standard (NPL India). Calibrate at intervals; after any shock or repair.

5. Limits, Fits and Tolerances

No two parts can be made **exactly** the same size - some variation is unavoidable. A **tolerance** is the permissible variation in a dimension. The designer specifies **limits** so that parts function and **interchange** .

Terminology (hole and shaft)

Term	Definition
Basic size (nominal)	size from which limits are derived; theoretical size, e.g. 40 mm
Actual size	size obtained by measurement
Limits of size	maximum (**upper limit**) and minimum (**lower limit**) permissible sizes
Tolerance	difference between upper & lower limits = **UL - LL**

Deviation	algebraic difference between a size and the basic size (upper / lower deviation)
Zero line	line representing the basic size; deviations +ve above, -ve below
Fundamental deviation	deviation nearest to the zero line (decides the position of tolerance zone)
MMC / LMC	maximum material condition (smallest hole, largest shaft) / least material condition
Allowance	intentional difference between the MMC of mating parts = min. clearance (+) or max. interference (-)

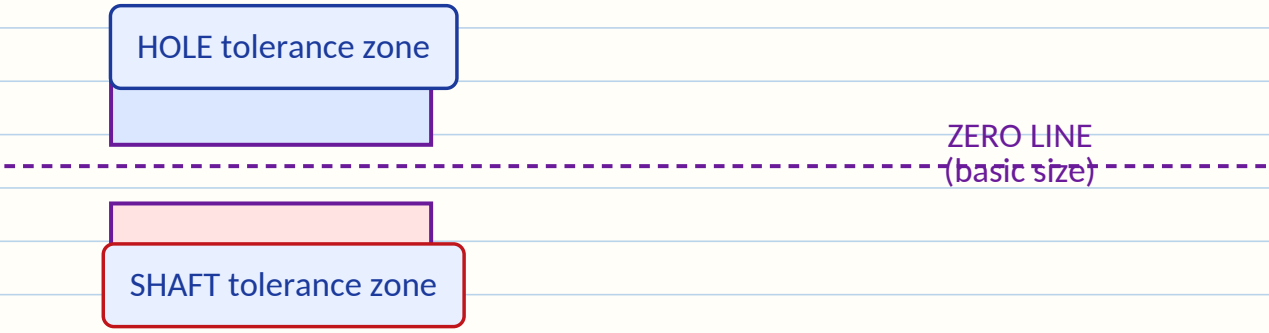


Fig 1.4: Tolerance zones of hole and shaft relative to the zero line

Types of tolerance

- **Unilateral** : tolerance in **one direction** only from basic size: 25 +0.02/-0.00.
  Bilateral : on both sides: 25 +-0.02 (equal) or +0.03/-0.01 (unequal).

Fits

Fit = relationship between the sizes of mating hole and shaft (before assembly).

1. **Clearance fit** : shaft always **smaller** than hole -> always a clearance. Free movement (bearings, sliding fits). E.g. H7/g6 (running), H8/f7.
2. **Interference fit** : shaft always **larger** than hole -> press / shrink fit; permanent assembly (bush in housing). E.g. H7/p6, H7/s6.
3. **Transition fit** : may be clearance or interference depending on actual sizes (location fit: keys, gear on shaft). E.g. H7/k6, H7/n6.

Fit	Clearance (max / min)	Condition
Clearance	Max C = HML hole - LML shaft; Min C = LML hole - HML shaft	Hole LL >= Shaft UL

Interference	Max I = HML shaft - LML hole; Min I = LML shaft - HML hole	Shaft LL >= Hole UL
Transition	Max C and Max I both exist	Tolerance zones overlap

Hole basis and shaft basis systems

Hole-basis system

Hole size is kept **constant** (fundamental deviation = 0, letter **H** : lower deviation zero) and the **shaft size varied** to get the different fits. **Preferred** because holes are produced by **fixed-size tools** (drills, reamers, broaches) and checked with plug gauges, while shafts are easily turned to any size.

Shaft-basis system

Shaft size is kept **constant** (letter **h** : upper deviation zero) and the hole size is varied. Used when **one shaft** carries many parts with different fits (cold-drawn bar, line shafting with pulleys, couplings).

ISO system of limits and fits

- **Tolerance grades** : **IT01, IT0, IT1 ... IT16** (20 grades). Lower number = finer. Gauges IT1-IT5, fits IT5-IT11, rough IT12-IT16. IT = i x multiplier where i (standard tolerance unit) = 0.45 D^1/3 + 0.001 D (microns, D = geometric mean of size range in mm).
- **Fundamental deviation letters** : holes **A...ZC** (capital), shafts **a...zc** (small). H/h = zero deviation. 28 positions.
- Designation: **40 H7/g6** -> basic size 40 mm, hole tolerance grade 7 with H position, shaft grade 6 with g position.

Interchangeability

- **Interchangeability** - parts made to the specified limits can be assembled or replaced **without selection, fitting or adjustment** . Basis of mass production, spares availability and low cost.
- **Universal (full) interchangeability** : any component fits any mate (screws, bolts, bearings). **Selective interchangeability / selective assembly** : parts are sorted into groups by size (piston-cylinder) - for very tight tolerances at low cost.

6. Limit Gauges and Gauge Design

A **limit gauge** checks whether a dimension lies **within its tolerance limits** (no numerical value is read). It has two members: **GO** end (checks the **MMC** ; must pass) and **NO GO / NOT GO** end (checks the **LMC** ; must not pass). Fast, cheap, usable by unskilled workers.

Types of gauges

Gauge	Use	Notes
Plug gauge	checks **holes** (diameter)	double-ended; GO plug long, NO GO plug short (shorter & chamfered); handle taper
Ring gauge	checks **shaft** diameter / threads	GO ring and NO GO ring (with groove on rim to identify)
Snap (caliper) gauge	checks **shafts** , widths, thickness	double-ended or progressive; adjustable snap gauges
Gap gauge	checks lengths between faces / thickness	fixed gap for large work
Taper gauges	taper plug & ring gauges	check taper & diameter of Morse tapers (contact blue)

Taylor's principle

Taylor's principle

- The **GO gauge** should check **all features** of the dimension **at the same time** (full form, e.g. a full-length cylindrical plug checks diameter, roundness, straightness) - the GO gauge must be **full form** .
- The **NO GO gauge** should check **only one feature at a time** (separate NO GO for each dimension; point contact) - so a defect in one feature is not masked by another.

- Reason: if a NO GO plug were full form, it might not enter because of an **ovality** error even when the hole size is over the upper limit - then a bad hole would be accepted. A short NO GO plug can enter at a large diameter and reject.

Design of GO and NO GO gauges

- **Gauge tolerance** : gauges themselves can't be made perfectly. Usually **10% of work tolerance** (IS 919: gauge tolerance depends on the grade). Tolerance zone

of gauges is **inside** the work tolerance zone so that no bad part is accepted.

- **Wear allowance** : GO gauges wear since they are used at every check - provide extra material = **5% of the work tolerance** (located between the work limit and the gauge tolerance zone). NO GO gauge needs **no** wear allowance (seldom enters).

Design example (Plug gauge for hole 25 +0.021 / 0)

Work tolerance = 0.021 mm; Gauge tolerance = 10% x 0.021 = **0.0021** ; Wear allowance = 5% x 0.021 = **0.00105** .

- GO plug: lower limit = 25.000 + 0.00105 = **25.00105** ; upper limit = 25.00105 + 0.0021 = **25.00315** (wear allowance applied inside the work tolerance).
- NO GO plug: upper limit = 25.021 (HML); lower limit = 25.021 - 0.0021 = **25.0189** .

- **Snap gauge for shaft** (e.g. 25 f8): GO snap = HML of shaft - wear allowance (and - gauge tol.), NO GO snap = LML + gauge tolerance (inside). Same logic mirrored for the shaft.

7. Linear Measurement

7.1 Vernier instruments

Based on the **vernier principle** : two scales of slightly different divisions - the main scale (MS) and sliding vernier scale (VS) - enable reading to a fraction of the smallest main division.

Least count of vernier

LC = 1 MSD - 1 VSD = (value of 1 main-scale division) / (number of vernier divisions)

- Example: 1 MSD = 1 mm; 50 VSD coincide with 49 MSD -> LC = 1/50 = **0.02 mm** .
- **Reading = MSR + (VSC x LC)** , where MSR = main-scale reading, VSC = coinciding vernier division.

- Types: **vernier caliper** (outside, inside, depth), **vernier height gauge** (with scriber; for marking & height on surface plate), **vernier depth gauge** , **gear-tooth vernier** (chordal thickness).
- Parts of caliper: fixed jaw with beam, sliding jaw, locking screw, fine adjustment, depth rod; **inside jaws** (knife edges).

- **Errors** : zero error, parallax, **Abbe's-principle violation** (jaws away from scale), measuring force, wear and jaw non-parallelism.

Example: MSR = 24 mm, 17th vernier division coincides, LC = 0.02 -> Reading = 24 + 17 x 0.02 = **24.34 mm** (zero error corrected).

7.2 Micrometers

Works on the **screw-nut principle** : when the screw is rotated one revolution through a fixed nut, it advances axially by one pitch. Pitch = 0.5 mm, thimble has 50 divisions -> **LC = 0.5/50 = 0.01 mm** .

- **Parts** : **frame (C-shaped)** , anvil, spindle, sleeve (barrel with datum line), thimble (circular scale), **ratchet stop** (constant measuring force), locknut.

Reading = Sleeve reading + (thimble division x 0.01) .

7.3 Slip gauges (gauge blocks)

End standards : rectangular blocks of **hardened steel / ceramic / carbide** , with two opposite faces **lapped flat and parallel** to a very high degree - used to set, calibrate other instruments, and for direct precise measurement. Invented by **C.E. Johansson** .

- **Wringing** : when two clean lapped faces are slid together with slight pressure and twist, they **adhere** by **molecular attraction + thin oil film** (force ~ 300 N). Size of the stack = sum of block sizes (wringing film ~ 0.00001 mm).
- **Grades** (BIS / IS 2984): **Grade 00** (reference / master, +-0.05 micron), **Grade 0** (inspection / calibration), **Grade 1** (workshop / toolroom), **Grade 2** (workshop). Standard sets: M87 (87 pieces), M112, M45, M33.
- Sizes: 1.001-1.009 (9 pcs, 0.001 step), 1.01-1.49 (49 pcs, 0.01 step), 0.5-24.5 (49 pcs, 0.5 step), 25-100 (4 pcs, 25 step) - M87 set.

Building a dimension: 41.125 mm (M87)

Choose from the **last decimal** to eliminate digits one by one:

- 41.125 - 1.005 = 40.12 -> - 1.02 = 39.1 -> - 1.1 = 38 -> - 38 (take 38 mm).
- **Blocks: 1.005 + 1.02 + 1.1 + 38 = 41.125 mm** (use as few blocks as possible).

7.4 Tool maker's microscope

A **versatile optical measuring instrument** for measuring **dimensions, angles, threads, profile and form** of small, delicate parts (cutting tool profiles, gauges, screw threads, templates) without contact.

- **Construction** : base with a **work table** (XY micrometer heads, 25 mm travel, least count 0.01 mm), glass plate, vertical **column** with microscope tube,

eyepiece with a cross-line reticle (rotating with angle scale 0-360 degrees, vernier 1 minute), illumination (lamp, filters).

8. Comparators

A comparator is a relative (comparative) instrument which compares the unknown dimension with a standard (slip gauge stack / master) and shows only the deviation on an amplified scale. It is not used for direct measurement .

Principle

Standard is set on the table -> comparator zeroed -> work replaces the standard -> pointer shows deviation (+/-).

- Uses : inspection of large-lot production , checking of parallelism, roundness, flatness, concentricity , calibrating gauges.

A. Mechanical comparators

- Dial indicator : plunger with rack moves a pinion -> gear train -> pointer. Magnification ~ 100-1000. Range 0-10 mm, LC 0.01 mm (0.001 mm). Stand with base and column.

B. Pneumatic comparators

Works on the change in air flow / back pressure as the gap between a nozzle (jet) and the work surface changes. Compressed air at constant pressure supplied through control orifice to a measuring jet .

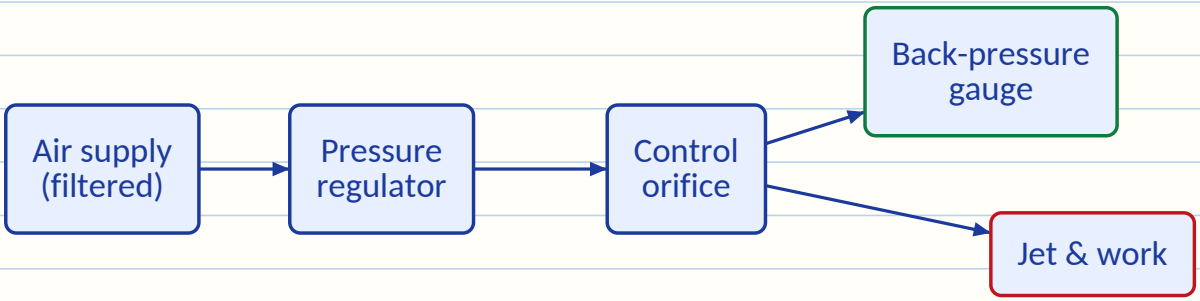


Fig 1.5: Solex back-pressure pneumatic comparator

- Back-pressure type (Solex) : back pressure P between orifice and jet changes with the gap; read by a manometer / dial. Linear between P1 and P2 region -> magnification up to 40,000 .
- Flow type : the flow rate through the gap is measured with a tapered glass tube (rotameter) - float height shows the gap.

C. Electrical and electronic comparators

- **LVDT** (Linear Variable Differential Transformer): plunger carries a **ferromagnetic core** between a primary and two secondary coils; displacement unbalances the secondaries -> output voltage proportional to displacement. Magnification 10^4-10^5; resolution 0.1 micron; supply AC.

9. Angular Measurement

Angles are measured in **degrees, minutes and seconds** (1 degree = 60', 1' = 60'') or **radians** . Instruments: **protractors, sine bar, angle gauges, clinometer, autocollimator, angle dekkor** .

9.1 Sine bar

A **precision** hardened and ground steel bar with **two cylindrical rollers** of equal diameter fixed at a **known centre distance L** (100, 200, 300 mm), the rollers' axes parallel to the bar's top face. Used to measure and set angles precisely using **trigonometry** and slip gauges.

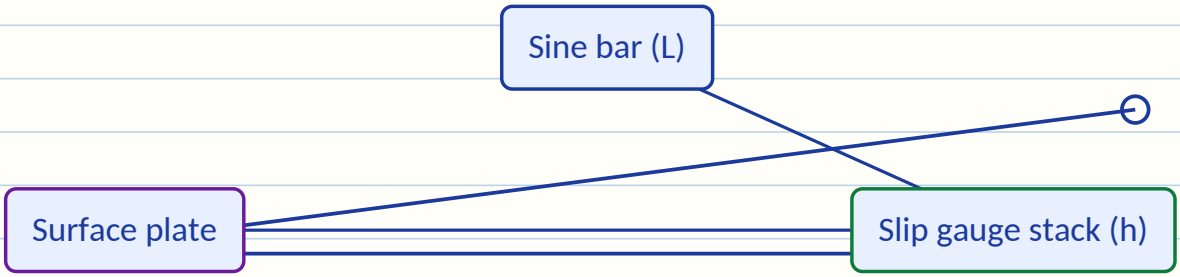


Fig 1.6: Sine bar - sin θ = h / L

Formula

sin θ = h / L -> h = L sin θ

- L = distance between roller centres; h = height of slip-gauge stack under one roller.

- **Example** : set a 200 mm sine bar to 30 degrees -> h = 200 x sin 30 = **100 mm** . For 25 degrees: h = 200 x 0.4226 = 84.52 mm.
- **Limitations** : **not suitable for angles > 45 degrees** (error in sine amplifies; as θ -> 90 degree, a small error in h gives a large angle error); **cannot be used for ... large parts** ; accuracy depends on slip gauges and roller diameters; **sensitivity decreases** at larger angles. Types: **sine centre** , **compound sine plate** , sine table, **sine vice** .

9.2 Bevel protractor (vernier)

- A **vernier bevel protractor** measures angles to **5 minutes (5')** . Parts: **stock (base)** , **blade** (sliding, adjustable), **dial (circular, 0-90-0 / 360 deg)** , **vernier scale** , lock nut, acute angle attachment.
- **Least count** : 12 vernier divisions = 23 main divisions (1 MSD = 1 degree). 1 VSD = 23/12 degree, so LC = 2 x 1 - 23/12 = 1/12 degree = **5 minutes** .
- Reading = Dial reading + (vernier coinciding division x 5'). Used for angle of **tool cutting edges, V-blocks, dovetails** , bevel gears.

9.3 Angle dekkor

An **optical, comparator-type** instrument (**autocollimator + angle gauges**) for measuring **small angular differences** ; makes use of **autocollimation** with an **illuminated scale** and a fixed **graticule** .

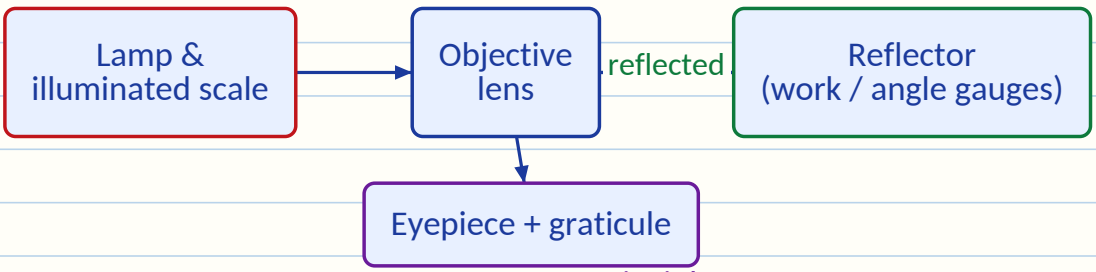


Fig 1.7: Optical principle of angle dekkor

- **Working** : Set the **slip-gauge combination** (angle gauges wrung) equal to the **nominal angle** on the surface plate - **reflector at that angle** ; read scale. Replace the angle gauge by the **work** ; the image shifts: **deviation read on the scale** (1 division = 1 minute; range ~ 1 degree). Image of the **illuminated scale** coincides with the graticule when the reflector is perpendicular to the optical axis.

9.5 Rollers and spheres for tapers

Taper measurement using precision rollers and spheres , **slip gauges** and a micrometer / height gauge - used for **taper plug gauges, ring gauges, dovetails, cones** .

Using two balls (spheres) in an internal taper (female)

Let R and r = radii of large and small spheres; H = depth difference measured to the tops of the balls (large above small); L = distance between ball centres = **H - R + r** .

sin (α/2) = (R - r) / L , where α = included angle of the taper.

Using rollers for a male taper (e.g. taper plug)

Two equal rollers (diameter d) placed on a plane at the large and small ends; measure the distance across them M1 and M2 over rollers at height difference H apart:

tan (α/2) = (M1 - M2) / 2H (rollers of same diameter placed at sections); then taper angle = α.

- **Example** : roller measurement: M1 = 68.4, M2 = 60.2 mm, H = 50 mm -> tan (α/2) = (68.4 - 60.2) / (2 x 50) = 0.082 -> α/2 = 4.69 degree; **α = 9.38 degrees** .

Quick Revision / Important Questions

1. Generalized measurement system; accuracy vs precision; errors and calibration.
2. Limits, fits, tolerances; hole vs shaft basis; Taylor's principle; design of GO / NO GO gauges.
3. Vernier caliper, micrometer, slip gauges (build 41.125 mm), tool maker's microscope.
4. Comparators (mechanical, pneumatic, electrical); sine bar, angle dekkor, bevel protractor; tapers with rollers & spheres.

Unit 2

Flatness & Surface Roughness

(Handwritten Notes)

Syllabus Topics Covered

1. Measurement of flatness: straight edges, surface plates
2. Optical flats, autocollimators, interferometers and their applications
3. Surface roughness terminology; roughness vs waviness
4. Numerical assessment: CLA, RMS, Ra, Rz
5. Methods of measurement: profilograph, talysurf
6. BIS symbols for indication of surface roughness

1. Flatness Measurement

A surface is **flat** when all its points lie in **one plane** . **Flatness error** = the **minimum separation of two parallel planes** between which the entire surface lies (**flatness tolerance zone**). Importance: machine-tool beds, slideways, surface plates, gauge faces, mating surfaces (leak-proof joints), optical parts.

Flatness tolerance

Surface must lie between **two parallel planes** a distance **t** apart (e.g. 0.02 mm).
Flatness controls **form** only, not orientation or position.

1.1 Straight edges

A **straight edge** is a **precision reference** with a long, **accurately flat** working edge, used to check the **straightness and flatness** of surfaces by comparing the work with the edge.

Type	Description / use
Toolmaker's (bevelled) straight edge	steel, bevelled edge ground & lapped; for small lengths (up to 300 mm); light-gap test
Wide-edge straight edge	wider face prevents tilting; checking of lathe beds, machine slides

I-section / box girder	cast iron or steel; rigid with minimum weight for lengths 1-3 m (machine tool beds)
Parallel-sided & angle straight edges	angle edge for dovetails and V-slides

- **Method 1 - Light-gap (light slit) method** : straight edge placed edge-down on the work with light behind; a **gap of 0.002 mm (2 micron)** is seen as a **white light** ; narrower gap shows blue, gap < 0.5 micron cannot be seen. Used to estimate error visually.
- **Method 2 - Engineer's blue (Prussian blue) method** : thin blue coat on the straight edge; slid over work; **high spots** show as blue patches - then scraped.
- **Method 3 - Feeler gauge / slip gauge** under edge to measure gap; **dial indicator** moved along the straight edge.
- Straight edge itself is checked by **reversal** or by comparing with a master.

1.2 Surface plates

A **surface plate** is a **flat, rigid reference plane** on which work is placed for inspection, marking-out, and as a datum for height gauges, dial indicators, sine bars, comparators and slip gauges.

- **Materials** : **cast iron** (close-grained, stress-relieved, ribbed, **scraped**), **granite** (black granite - dense, non-magnetic, **no burrs, rust-free** , low thermal expansion, hard, stable), **glass** , **ceramic** .
- **Grades (IS 2285) : Grade 00 (laboratory master) = AA , Grade 0 = A (inspection room) , Grade 1 = B (toolroom/workshop)** , Grade 2 (rough). Flatness tolerance for grade 1 (1000 x 1000 mm): ~ 10 micron.

Cast-iron plate	Granite plate
Rusts; burrs raised on scratches	Rust-free; scratch makes only a crater - no burr
Magnetic; warps with temperature	Non-magnetic; low thermal expansion
Cheaper; threaded holes & T-slots possible	Costly; brittle; chips on impact; inserts needed
Needs periodic re-scraping	Easier maintenance; wears evenly; good damping

- **Making** : cast, seasoned (aged), rough-planed, ground, then **hand-scraped** to flatness using three-plate method (Whitworth) with Prussian blue; **lapped** for

high grades. Supported at **three points (Airy points)** to avoid sagging.

- **Checking flatness of a surface plate** : (i) **Autocollimator** ; (ii) **spirit-level / precision level** on a grid; (iii) **Moody's method** with a ... (iv) **laser interferometer** .

1.3 Spirit level / precision level

- A **sensitive bubble vial** in a rigid body; the bubble moves when the base tilts. Sensitivity: **0.02 mm/m** (about 4 seconds of arc per division). Readings along **line (Moody) grid pattern** (the Union Jack pattern) give relative heights; the **plot** of heights shows flatness.

1.4 Optical flat

An **optical flat** is a **disc of fused quartz, glass or Pyrex** (diameter 25-300 mm, thickness ~ 1/6 of diameter) with **one or both faces lapped and polished** to a flatness of 0.05 micron (**1/20 of wavelength**). Used to check the **flatness of gauge faces, slip gauges, micrometer anvils** , surface plates.

- **Principle** : **interference of light** - when the flat is placed at a **small angle (wedge)** on the surface, **monochromatic light** reflected from the **bottom of the flat** and from the **work surface** interferes: **alternate dark and bright bands (fringes)** are formed.

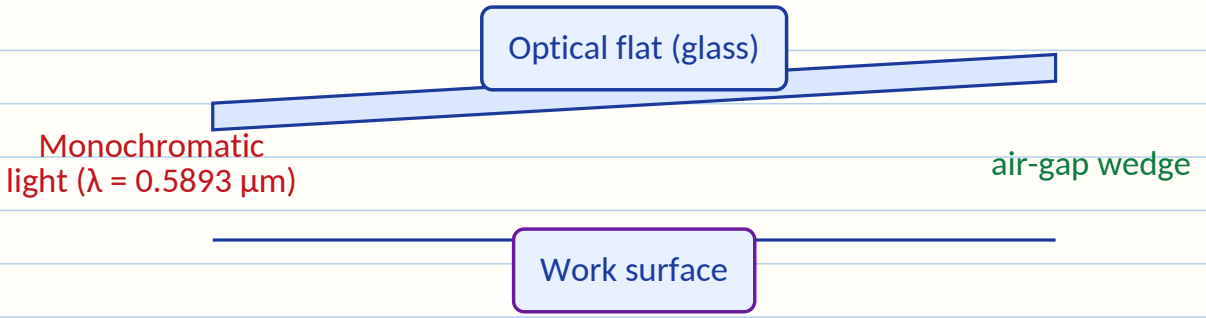


Fig 2.1: Air-wedge between optical flat and work

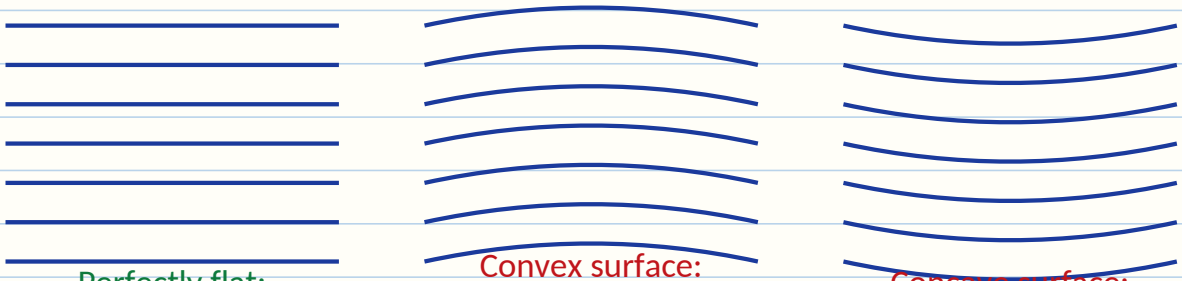
Key relation

Each **fringe** = a change in air-gap of **λ/2** .

- For monochromatic helium light λ = 0.5893 micron -> **λ/2 = 0.2945 micron (about 0.3 micron)** per fringe.
- **Flatness error = (number of fringes of curvature) x λ/2** .

- **Reason** : the extra path of the ray going through the air gap is 2t; **dark band** where 2t = nλ (with 180 degree phase change at reflection from the denser surface); **bright band** where 2t = (n + 1/2)λ.

Interpretation of fringe patterns



Perfectly flat: straight, parallel, equispaced

Convex surface: fringes curve toward the line of contact

Concave surface: fringes curve away

- **Curved fringes** : the **deviation of a fringe from straightness / spacing between fringes** = amount of departure. If a fringe bends by **a** while the **spacing is b** , **flatness error = (a / b) x λ/2** .
- **Closed circular rings (Newton's rings)** -> surface is **spherical** (convex or concave); number of rings x 0.3 micron = depth. **Saddle / cylindrical** surfaces show **parabolic bands** .
- **Wringing** a good optical flat on the gauge face -> fringes absent (full dark/ uniform) for an extremely flat surface.
- **Care** : clean, handle with soft gloves/tissue, **never slide** (scratches), store with cover; **check the flat** itself by comparing with a master flat (**three-flat method**).

2. Autocollimator

An **optical instrument** used to measure **small angular deviations** , **flatness, straightness, squareness and parallelism** . It is a combination of a **telescope and a collimator** : it projects a **parallel beam** , receives the **reflected beam** from a **mirror** (placed on the object) and measures the **tilt of the mirror** .

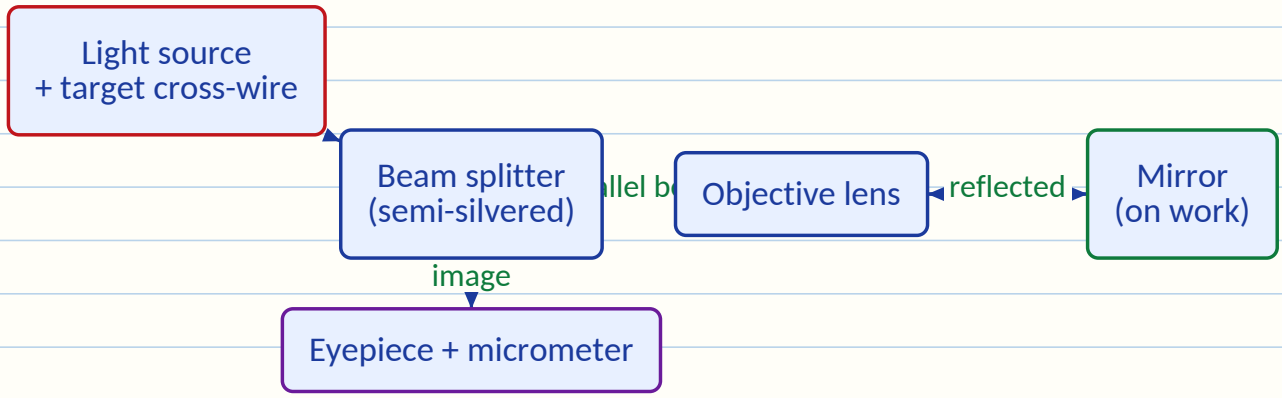


Fig 2.2: Principle of the autocollimator

- **Principle** : if the mirror is **exactly perpendicular** to the optical axis, the reflected image of the target falls on the **target itself** (coincides with the cross-wire). If the mirror **tilts by θ** , the reflected beam deviates by **2θ** , and the image is displaced by **x = 2 f θ** (f = focal length of the objective).

Basic relation

x = 2 f θ -> θ = x / (2 f)

- The displacement x is read on the **micrometer eyepiece** (1 division = 1 second of arc / 0.1 second in precise models). Range ~ 10 minutes; sensitivity 0.1 - 1 second.

- **Types** : visual (eyepiece), **digital / electronic** (CCD, photo-electric), **laser autocollimator** , **dual-axis** .

Use for flatness testing of a surface plate

1. Divide the surface into a **grid of lines** (Moody / Union Jack pattern) with the line length equal to the **base length of the mirror carriage** (e.g. 100 mm).
2. Place the **reflector (mirror) on a base** at one end; autocollimator is fixed at the other end - read the **angle** (in seconds) at each successive position.
3. Each reading gives the **tilt** of the segment; **height change** between ends of base b = **b x θ** (θ in radian: 1 second = 4.848 x 10^{-6} rad).
4. **Cumulate heights** along the line to obtain **profile** , repeat for all lines, then find the **minimum two-plane zone** -> flatness error.

- **Example** : base length 100 mm, readings 5, -3, 4 seconds -> rise = 100 x 5 x 4.848 x 10^{-6} = **0.0024 mm** for first step.
- **Applications** : flatness & straightness of **machine-tool slideways** , surface plates; **squareness** of two surfaces (with optical square / pentaprism);

parallelism ; angular measurement (along with angle dekkor); alignment of bearing bores and shaft lines ; checking small angles of tilt in machine movement (pitch, yaw).

3. Interferometers

Interferometry = the technique of using interference of light waves to measure lengths, flatness and displacement with accuracy up to a fraction of the wavelength of light . The interferometer splits a light beam into two paths, recombines them and the fringe pattern shows the path difference .

- Source requirements : monochromatic and coherent (sodium, mercury-198, cadmium, krypton-86, He-Ne laser).

3.1 NPL flatness interferometer

- Developed at National Physical Laboratory (UK) ; for calibration of slip gauges, end bars (flatness, parallelism, length) in terms of wavelength.
- Construction : mercury-vapour lamp (green light 0.5461 micron) -> condenser, slit, collimating lens -> a dispersing prism to separate the wavelengths (selects a single wavelength), then the beam -> optical flat placed over the gauge block wrung on a base plate . The reflected rays from the base plate and from the gauge block top give two sets of fringes (offset) viewed through the telescope .
- Measurement : the fringe displacement (a) relative to fringe spacing (b) gives the fractional order: length = (n + a/b) λ/2 . Using several wavelengths (red, green, blue cadmium) and the method of exact fractions the integral order n is found.

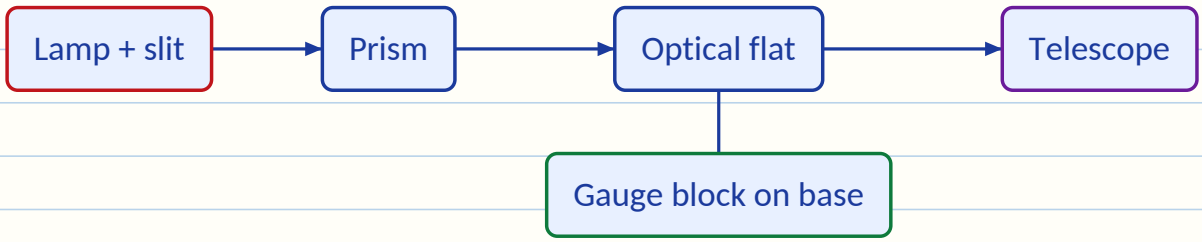


Fig 2.3: NPL flatness interferometer (simplified)

3.2 Michelson interferometer / laser interferometer

- Michelson : a beam splitter divides the beam; one goes to a fixed mirror , the other to a moving mirror (on the machine slide). When the moving mirror moves by λ/2 , the path difference changes by λ , so one fringe passes.

Counting fringes gives displacement: $d = N \lambda / 2$.

- Laser interferometer (Hewlett-Packard type): He-Ne laser (λ = 0.633 micron), stabilized ; retro-reflectors and photodetectors with reversible counters ; resolution 0.01-0.1 micron ; range up to tens of metres ; with environment compensation (air temperature, pressure, humidity).
- Applications : calibration of machine-tool positioning , CMMs , slip gauges , angular (with optics), straightness , flatness , squareness ; vibration & displacement measurement; length standard realization; optical component testing.
- Advantages : extremely high accuracy, non-contact, traceable to wavelength. Disadvantages : costly, sensitive to vibration and air disturbances , needs skilled operators.

Instrument	Measures	Typical resolution
Straight edge + feeler	straightness, flatness	0.002 mm
Spirit level	inclination, flatness	0.02 mm / m
Autocollimator	small angles, flatness	0.1 - 1 arc-second
Optical flat	flatness of small faces	0.3 micron / fringe
Laser interferometer	length, straightness, flatness	0.01 - 0.1 micron

4. Surface Roughness Measurement

No machined surface is perfectly smooth - it has irregularities left by the manufacturing process. Surface texture = repetitive or random deviations from the nominal surface - includes roughness, waviness, lay and flaws . It affects friction, wear, lubrication, fatigue strength, corrosion resistance, fits, appearance and heat transfer .

Surface profile = roughness (fine, blue) + waviness (green)

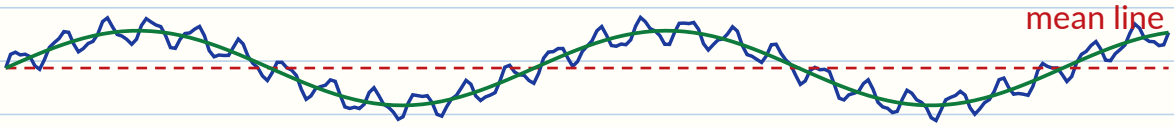


Fig 2.4: Roughness and waviness on a surface profile

Terminology (IS 3073 / ISO 4287)

Term	Meaning
Actual surface / profile	surface (trace) as it really exists - cut by a plane perpendicular to the surface
Roughness	**closely spaced** finer irregularities produced by the **cutting tool / grains of abrasive** , feed marks
Waviness	**widely spaced** irregularities (greater spacing than roughness) due to **machine vibration, chatter, deflection of work, warping, heat treatment**
Lay	direction of the **predominant surface pattern** , usually decided by the production process
Flaws	irregularities that occur **at one place / random** : scratches, cracks, blow-holes, inclusions
Sampling length (cut-off) l	length of profile used for assessing roughness: 0.08, 0.25, 0.8, 2.5, 8, 25 mm. Waviness is filtered out by the cut-off.
Evaluation length	length of profile used for assessment; normally **5 sampling lengths**
Mean line (M-system)	line so drawn that, within the sampling length, the **sum of areas above = sum of areas below** it
Peak / valley	highest point / lowest point of profile above / below the mean line
Primary profile	total profile (roughness + waviness + form) after removing nominal form

Differences between roughness and waviness

Surface roughness	Surface waviness
Closely spaced, fine irregularities	Widely spaced, longer wavelength irregularities
Pitch (spacing) **< 0.8 mm** approx.	Spacing **0.8 - 10 mm** (roughly 1000x height)
Caused by **cutting action** - tool feed, tool tip, abrasive grains, built-up edge	Caused by **vibration, chatter, machine deflection, heat treatment, clamping strain**
Height measured in **microns** ; Ra, Rz	Height measured as **Wt / Wa** (larger amplitude)
Superimposed on waviness	Superimposed on form errors

- **Typical order of magnitude of roughness: lapping 0.05-0.4 micron Ra; honing 0.1-0.8; grinding 0.1-1.6; reaming 0.4-3.2; turning/milling 0.8-6.3; shaping/planing 1.6-12.5; sand casting 12.5-25; forging 3.2-12.5** .

5. Numerical Assessment of Surface Finish

Surface finish is numerically assessed by **parameters** computed from the profile **within the sampling length L** relative to the **mean line** ; y = height of the profile above (+) / below (-) the mean line at distance x.

1. Ra - Centre Line Average (CLA) / Arithmetic average

Ra = (1/L) $\int_0^L |y|\,dx$ (approx. **Ra = (|y1| + |y2| + ... + |yn|) / n**)

- Ra = **arithmetic mean of absolute deviations** of the profile from the mean line. Most widely used; unit **micron (µm)** .
- Also called **AA (arithmetic average)** or **CLA (centre line average, British)** .

2. Rq - RMS value (root mean square)

Rq = sqrt[(1/L) $\int_0^L y^2\,dx$] ≈ sqrt[$(y1^2 + y2^2 + ... + yn^2) / n$]

- More sensitive to peaks and valleys than Ra. **For a sine wave: Rq = 1.11 Ra** (approx. relation **RMS = 1.11 CLA**). Used in USA (RMS in micro-inch).

3. Rz - Ten-point height of irregularities

Rz = [(R1 + R3 + R5 + R7 + R9) - (R2 + R4 + R6 + R8 + R10)] / 5

- (average of the **five highest peaks** minus average of the **five deepest valleys**) in the sampling length, measured from a line parallel to the mean line. (DIN Rz = mean of 5 successive Rmax.)

Parameter	Name / description
Rmax (Ry, Rt)	**maximum peak-to-valley height** within the sampling (evaluation) length
Rp / Rv	maximum **peak height** above / maximum **valley depth** below the mean line
Rsm	mean **spacing** of profile elements (peaks) - lateral parameter
Rmr (tp)	**material ratio** (bearing ratio) at a given depth - Abbott-Firestone curve; predicts wear-in & load carrying

Example : profile ordinates (micron): +3, -2, +4, -5, +2, -4 -> Ra = (3 + 2 + 4 + 5 + 2 + 4)/6 = **3.33 micron** ; Rq = sqrt[(9 + 4 + 16 + 25 + 4 + 16)/6] = sqrt(12.33) = **3.51 micron** .

- **Why Ra alone is insufficient** : two different profiles (sharp peaks vs. wide plateaus) can have the same Ra - hence Rz, Rmax, Rsm and Rmr are used with it.
- **Roughness grade numbers (N-series, ISO 1302)** : N1 = 0.025 micron Ra, N2 = 0.05, N3 = 0.1, N4 = 0.2, N5 = 0.4, N6 = 0.8, N7 = 1.6, N8 = 3.2, N9 = 6.3, N10 = 12.5, N11 = 25, N12 = 50 micron.

6. Methods of Measurement of Surface Finish

Method	Examples	Remarks
Comparison (qualitative)	touch, visual, **fingernail** , **surface-roughness comparison specimens** (standard plates for turning, milling, grinding), scratch/viewing with microscope	rough, subjective, quick - shop floor
Direct (stylus)	**Talysurf** , profilometer, **profilograph** (surface analyzer)	quantitative; Ra, Rz, profile record

Optical	light-section microscope (Schmaltz) , interference microscope (Tolansky) , double-slit, speckle	non-contact; for very smooth surfaces
Other	pneumatic, capacitance, scanning probe (AFM, SEM)	nano scale

6.1 Stylus (Talysurf / profilograph) method

A diamond stylus (tip radius 2-10 micron, 90 degree cone) is traversed across the surface ; the vertical motion of the stylus (caused by roughness) is converted by a transducer into electrical signals, amplified, filtered and given as a profile (graph) or Ra reading .

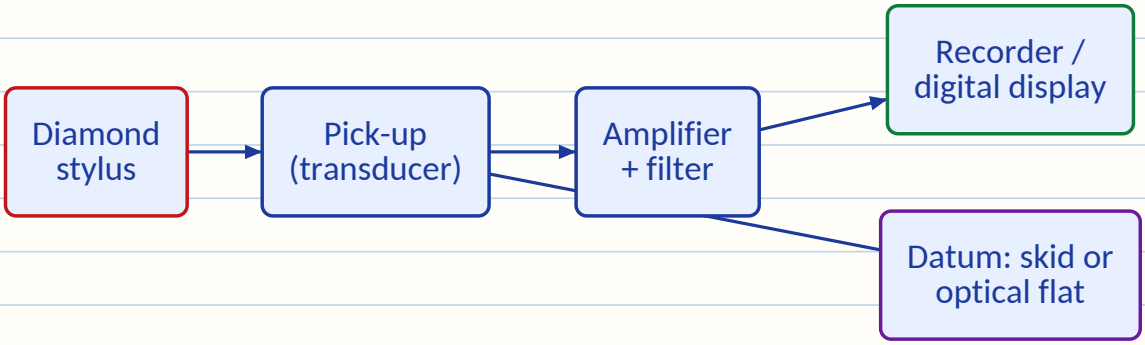


Fig 2.5: Block diagram of Talysurf

- Talysurf (Taylor-Hobson): the stylus is attached to an arm pivoted on a knife edge ; the other end carries an armature between two E-shaped coils (inductive pick-up, differential). Roughness moves the armature -> unbalance of the AC bridge -> modulated signal -> demodulated, amplified -> recorded; Ra computed electronically.
- Datum : skid (a blunt, long-radius shoe riding on the surface; the skid follows the waviness - acts as a mechanical filter), or an independent datum (optical flat / precision slide) when waviness has to be measured.
- Drive unit : motor driving the pick-up at a constant traverse speed (e.g. 1 mm/s); traverse length = evaluation length; cut-off filter (0.25, 0.8, 2.5 mm) (electrical 2RC filter / Gaussian).
- Magnification : vertical 500 - 100,000x; horizontal 10 - 1000x. Measuring range 0.01 - 100 micron.
- Limitations : stylus radius can't enter very fine valleys; stylus pressure may

scratch soft surfaces; vibration of the shop; slow .

6.2 Profilograph / Profilometer

- **Profilograph** (Surface analyzer): a **recording** instrument that **draws the magnified profile** (like a chart) of the surface; the stylus movement is amplified electronically and **plotted** on paper.
- **Profilometer** : a **meter-type** instrument (Brush Surfindicator, Perthometer) - the stylus **traces** the surface and a **pointer** (or digital display) shows **Ra** or RMS **directly** , without graph; **fast** , convenient on the shop floor and **portable** .

6.3 Optical methods

- **Light-section (Schmaltz) microscope** : a thin **slit of light** at **45 degree** to the surface forms a **bright line image** that follows the profile; viewed through the microscope at **90 degree** - magnified profile (0.8-80 micron).
- **Interference microscope (Tolansky)** : optical flat over the surface; **fringes** displaced by the irregularities - **fringe displacement = λ/2** per fringe; for **very fine finishes** (< 0.025 micron).

7. BIS Symbols for Indication of Surface Roughness

IS 696 / IS 3073 / IS 10719 (ISO 1302) : standardized symbols on **engineering drawings** to specify the **surface finish** and **machining** requirements.

Symbol	Meaning
Basic symbol (two unequal legs, 60 degree)	surface **texture may be obtained by any method** ; alone it has no meaning without further information
Basic symbol + **horizontal bar** (closed triangle)	**machining required** (material removal necessary - cutting, grinding, etc.)
Basic symbol + **circle**	**material removal prohibited** - surface left as from casting, forging, rolling, etc.
Symbol + **circle on the closed line**	same surface finish required on **all surfaces** around the contour

Position of specifications on the symbol:

a = Ra value (µm) or N grade	b = production method, treatment, coating	c = sampling length (cut-off, mm)
d = lay direction symb	e = machining allowance (mm)	f = other roughness (Rz)

Lay symbols

Symbol	Direction of lay	Example
=	**parallel** to the plane of projection of the view	shaping, planing
⊥	**perpendicular** to the plane of projection	shaping (ends), turned face
X	**crossed** in two slant directions	honing
M	**multi-directional** (no particular direction)	lapping, grinding, super-finishing
C	approximately **circular** relative to the centre	face turning
R	approximately **radial** relative to the centre	end milling

Indication on drawings

- The symbol is drawn **on the outline of the surface or on an extension line** , **pointing to the surface from outside the material** (like a dimension) and normally **read from the bottom or right** side of the drawing.
- Example: **Ra 1.6** (or **N7**) written above the symbol = roughness average **1.6 micron** ; **max** and **min** values can be given (e.g., Ra 3.2 max, Ra 1.6 min). The sampling length is given below the horizontal bar (e.g., **0.8**); machining allowance **2 mm** .
- **Common finishes / uses** : **N12-N10** rough surfaces (non-mating); **N9-N8** medium (general machining); **N7-N6** fine (mating parts, bearings seats); **N5-N4** very fine (piston, journals, **slideways**); **N3-N1** super-finish (gauges, lapped, **slip gauges**).

Quick Revision / Important Questions

1. Flatness: straight edges, surface plates, optical flat - principle and fringe patterns.

2. Explain the autocollimator principle and its use for flatness.
3. Interferometers (NPL, laser) and their applications.
4. Roughness vs waviness; Ra, RMS, Rz formulas with examples.
5. Talysurf and profilograph; BIS symbols and lay.

Unit 3

Screw Thread & Gear Measurement, CMM

(Handwritten Notes)

Syllabus Topics Covered

1. Elements of screw threads; errors in screw threads
2. Measurement of external and internal threads
3. Screw thread gauges
4. Gear tooth terminology
5. Measurement of gear elements: run-out, lead, pitch, backlash, profile, pressure angle
6. Tooth thickness, diameter, constant chord and base tangent methods
7. Coordinate Measuring Machine (CMM): construction and features

1. Screw Thread Measurement

1.1 Elements (terminology) of a screw thread

Basic profile of an ISO metric (60 degree) thread

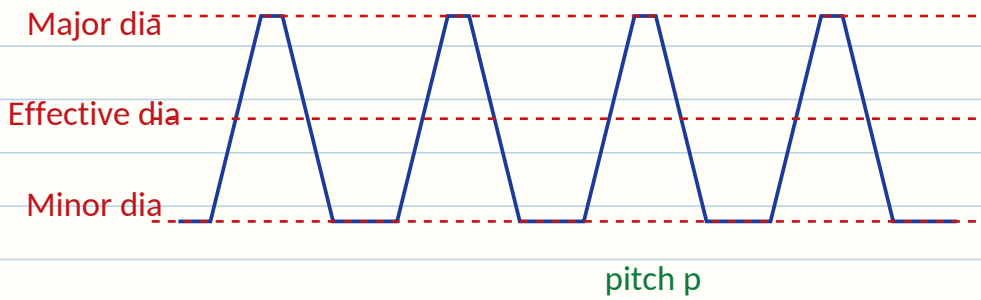


Fig 3.1: Thread profile with major, effective (pitch) and minor diameters

Term	Definition
Major (outside) diameter	largest diameter of a thread (crest of external, root of internal); **nominal size**

Minor (core / root) diameter	smallest diameter of the thread (root of external, crest of internal)
Pitch (effective / simple effective) diameter	diameter of an **imaginary co-axial cylinder** which cuts the thread so that the **width of the groove = width of the thread** (at that cylinder)
Pitch (p)	distance from a point on one thread to the corresponding point on the **next thread** , **parallel to the axis**
Lead (L)	axial distance **advanced in one revolution** : **L = n x p** (n = number of starts)
Flank angle / thread angle	angle between flank and the perpendicular to the axis / included angle between the two flanks (60 deg metric; 55 deg Whitworth)
Crest & root	top surface joining two flanks / bottom surface between two flanks
Depth of thread	distance between crest and root measured **perpendicular to the axis**
Helix angle	angle made by the helix of thread at the pitch cylinder: **$\tan \psi = L / (\pi d_2)$**
Flank	straight surface connecting crest and root

Thread form	Angle	Use
ISO metric	60 degree (flat crest)	fasteners, general
Whitworth (BSW)	55 degree (rounded)	British pipe & fasteners
ACME	29 degree	lead screws, power transmission
Square	0 degree flanks	vices, jacks (high load, low friction)
Buttress	45 / 7 degree	one-direction heavy load (press, jacks)

1.2 Errors in screw threads

1. **Pitch errors** : (a) **Progressive** - pitch uniformly longer or shorter than nominal over the length (error in lead screw of lathe; **tool-feed** or **expansion** errors). (b) **Periodic** - repeats at regular intervals per revolution (eccentric mounting of blank / **gear errors** in the lead-screw drive). (c) **Drunkenness** - the thread form

is a **helix with a periodic variation** : axial section looks like straight but the **thread curve is not perfect helix** (cutting tool inclined). (d) **Irregular** - due to machine, tool wear, material variations.

2. **Major / minor diameter errors** : cause interference with mating parts at the root/crest.
3. **Effective diameter errors** - most critical: affect the fit (due to wrong depth of cut).
4. **Flank angle errors** : **angle error** (angle of both flanks differ from nominal) or **unequal flank angles / bisector not perpendicular to axis** (tool setting).
5. **Form errors** : wrong crest/root radius, **rounded** or irregular profile.

Effect of pitch / angle error

A pitch error or flank-angle error **virtually increases the effective diameter** (of an external thread):

- **Virtual effective diameter = measured effective diameter + (compensation for pitch error P + flank angle error)**
- Pitch error compensation: **1.732 |ΔP|** , Angle compensation: **0.0003 p (Δα in minutes)** (metric 60 degree).

1.3 Measurement of external screw threads

(a) Major diameter

- **Bench micrometer** (or ordinary **micrometer** with flat anvils) measures the outside diameter over the crests; **bench micrometer** with **standard cylinder** gives comparison with a **setting cylinder** of equal size, with constant pressure (fiducial indicator). Accuracy 0.002-0.005 mm.

(b) Minor diameter

- Measured with a **micrometer** fitted with **tapered V anvils / small radius pieces (prism pieces)** touching the root of the thread. Or **toolmaker's microscope** .

(c) Effective (pitch) diameter

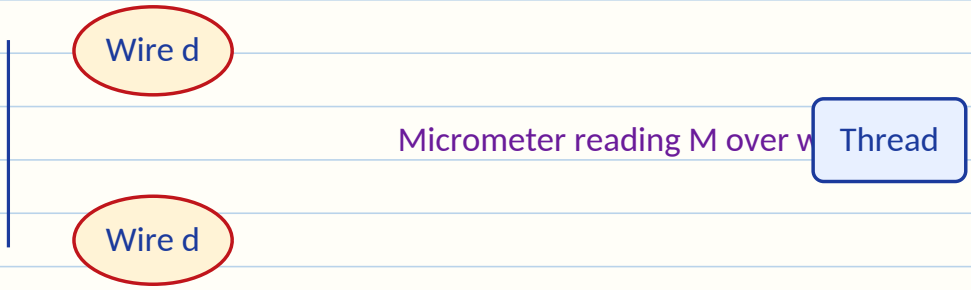
1. Screw thread micrometer - the anvil is a **V** (60 deg) and spindle a **cone** that fit the thread flanks; reading gives the effective diameter directly (error: pitch error of the micrometer; need set for each pitch range).

2. Two-wire method : two **accurately ground wires of same diameter d** are placed in **opposite grooves** and **measured over the wires by a micrometer** (M). Third wire is not used.

Two / Three wire formulae (metric 60 degree thread)

Effective diameter E = M - P where **P = d (1 + cosec (α/2)) - (p/2) cot (α/2)**

- For **metric** (α = 60): **P = 3d - 0.866 p** ; For **Whitworth** (α = 55): P = 3.166 d - 0.9605 p.
- **Best wire size** (touches the flanks on the **pitch line**): **d = p / (2 cos (α/2))** = **0.5774 p** (metric) = 0.5637 p (Whitworth).
- **Over the wires: M = E + 3d - 0.866 p** (metric).



- **Example** : M10 x 1.5 thread, best wire d = 0.5774 x 1.5 = **0.866 mm** ; E = 9.026 mm -> M = 9.026 + 3 x 0.866 - 0.866 x 1.5 = 9.026 + 2.598 - 1.299 = **10.325 mm** .

3. Three-wire method : **three wires of same diameter** - two on one side and one on the opposite side; the micrometer is placed over the **single wire and the pair** - **more accurate** than two-wire as it **aligns the axis** of the micrometer perpendicular to the thread axis, and cancels the effect of inclination.

4. Floating carriage micrometer (Diameter measuring machine) : the thread is mounted **between centres** ; carriage (with a micrometer and **fiducial indicator**) floats freely on ball slides. A **setting cylinder** of known diameter and wires are used: **E = (cylinder effective size) + (difference of readings: thread - cylinder)** (comparison method). Used also to measure **major & minor diameters** and **pitch** .

(d) Pitch measurement

- **Pitch measuring machine** : the thread is **mounted between centres** ; a **stylus with a ball** (or a conical point) is brought into the successive grooves and the carriage advance is read with a **micrometer** - **cumulative pitch error** and **progressive** or **periodic** errors can be seen. **Screw pitch gauge** (set of leaves) gives **quick comparison** of the pitch.
- **Toolmaker's microscope** (also **optical projector**): pitch measurement along axis

with the cross hair at the corresponding points on the successive threads.

(e) Thread flank angle & form

- **Toolmaker's microscope** (angular scale in the eyepiece, crosshair aligned with flank) - angle by rotating the eyepiece, **profile projector** with **master chart (overlay)** ; **optical comparator** (shadow method) - simple and fast.

2. Measurement of Internal Threads

More **difficult** than external threads because the thread is **inside** a hole, and a standard micrometer cannot reach.

- **Major & minor diameters** : **taper parallel (plug) setting** with a micrometer; **internal micrometer** , **telescopic gauge** , **ball-end** inside caliper; the **minor diameter** is checked by **GO / NO GO plain plug gauge** , and the **major** by thread plug.
- **Effective diameter** : **with taper parallels and rollers (balls)** - a pair of **taper parallels** with **three rollers** etc., so the reading over the rollers is taken with a micrometer; alternatively **ball method** (balls of best size inserted in the groove and **depth micrometer / height gauge** measures the distance across).
- **Casts (impression)** : take a **cast of the internal thread** (plaster / metal / resin cast) and measure the cast as an external thread.
- **Pitch & angle** : **toolmaker's microscope** after **sectioning** , or by **impressions** ; **optical** (using a prism-and-microscope arrangement) via **bore scope** .
- **Thread plug gauges** (GO / NO GO) are the most **practical** shop floor inspection method.

Screw thread gauges

Thread gauges (limit gauges) check the **conformity** of thread within limits **quickly** , according to **Taylor's principle** .

Gauge	Checks	Details
Thread plug gauge	internal thread (nut)	**GO** : **full form** , long (engages full length) - checks **effective dia + major + minor** . **NO GO** : **truncated flanks** , **short** (2-3 threads) - checks **effective dia only** ; **minor diameter** checked by plain plug
Thread ring gauge	external thread (bolt)	GO ring: full form; **NO GO ring** : truncated thread, shorter. **Adjustable ring gauges** for wear

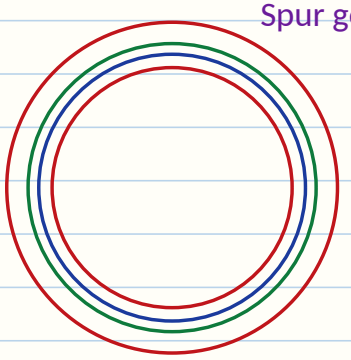
Caliper (snap) thread gauge	external thread	two pairs of rolls / anvils; GO rolls full form; NO GO rolls flank contact only
Screw pitch gauge	pitch (identification)	set of **steel leaves** with thread forms; select the leaf that fits
Setting plugs / check plugs	inspection of ring gauges	**setting plug** to set adjustable gauges; **check plug** to confirm wear of the GO gauge

- **NO GO is truncated** (flank contact only at the pitch diameter) so a **fault in major/minor diameter** doesn't make it pass/reject wrongly; thread NO GO gauge **must not enter more than 2 turns** .
- **Tolerance classes (IS 4218)** : 6H (internal medium), 6g (external medium); **fine / medium / coarse** classes (4H5H, 6H, 7H; 4h, 6g, 8g). **Designation** : **M10 x 1.5 - 6H** .

3. Gear Measurement

3.1 Gear tooth terminology (spur gear)

Spur gear tooth terminology



Addendum circle

Pitch circle

Dedendum (root) circle

Base circle

Term	Meaning / formula
Pitch circle (diameter D)	imaginary circle that **rolls without slipping** with the pitch circle of the mating gear
Module (m)	**m = D / N** (mm) - pitch diameter divided by number of teeth (metric standard)
Diametral pitch (P)	**P = N / D** (teeth per inch); **m x P = 25.4**
Circular pitch (p_c)	distance between corresponding points of adjacent teeth along the pitch circle: **p_c = π m = π D / N**
Addendum (a)	radial height of tooth **above** the pitch circle (= **1 m**); **Dedendum (d)** = below the pitch circle (= **1.25 m**)

Clearance (c)	dedendum - addendum = **0.25 m**
Pressure angle (φ)	angle between the **line of action** (common normal) and the **common tangent** to pitch circles: **14.5 or 20 degree** standard
Base circle	circle from which the involute is generated: **$D_b = D \cos \varphi$** ; **base pitch = $\pi m \cos \varphi$**
Tooth thickness (t)	arc thickness at pitch circle = **$\pi m / 2$ = 1.5708 m**
Backlash	clearance between non-working flanks of mating teeth along the pitch circle
Face width, lead (helical), whole depth	width of tooth parallel to axis; whole depth = **2.25 m** ; **outside dia = m (N + 2)**

Errors in gears

- **Profile (involute) error** , **pitch error** (**single/adjacent** , **cumulative**), **lead error** (helical), **radial run-out** (eccentricity), **tooth thickness** error, **backlash** , **alignment (parallelism / skew) error** , **composite error** (**total & tooth-to-tooth**), **surface finish** .
- **Causes** : errors in the **gear cutting machine** (**hob** , indexing/worm gear), **improper mounting** of blank, **tool wear** , **heat-treatment distortion** and **misalignment** .

4. Measurement of Gear Elements

4.1 Run-out (radial eccentricity)

Run-out = variation in the **radial position** of the **pitch line** with respect to the **axis of rotation** , due to **eccentric mounting** or an **inaccurate** gear blank.

- **Method** : gear is **mounted between centres** (or on an **accurate mandrel**); a **dial indicator** (with a **ball or cylindrical probe / roller** of suitable diameter) is placed in **successive tooth spaces** - **the pointer variation (Max - Min)** = run-out. Plotting gives **eccentricity** .

4.2 Pitch measurement

- **Pitch error** = difference between the actual and theoretical pitch. **Adjacent (single) pitch error** and **cumulative pitch error** (sum over a sector or full circle).
- **Pitch measuring instrument** (**Gear pitch comparator**): has a **fixed finger** and **adjustable finger** touching two successive teeth at the pitch circle; a **dial indicator** shows the **deviation** from the **set master pitch** (set using slip gauges). Both **base-pitch** and **circular-pitch** measurement.

- **Base pitch (normal pitch) measuring : base-pitch gauge (Ludwig type)** with **two fixed contacts (pre-set)** touching adjacent teeth flanks **tangentially** along the **line of action : base pitch $p_b = \pi m \cos \varphi$** (module 4, 20 degree: 11.81 mm).
- **Using optical dividing head / indexing table** with a **dial indicator - each tooth gap** is checked at successive angular positions (**360/N**).

4.3 Lead measurement (helical gears)

- **Lead** is the axial distance through which a **helical tooth** advances in one turn. **Lead checking instrument** : gear on **centres** with **a probe contacting the tooth flank** ; **rotate the gear** while the **probe moves axially (helical guide / sine bar guide)** at the **set lead** ; the **dial indicator** shows **lead deviation** across the face width.
- For **spur gears** the **lead error = parallelism error** of tooth traces to the axis, checked by dial indicator moving parallel to the axis.

4.4 Backlash

- **Backlash = max. tooth space - min. tooth thickness** ; **needed for lubrication & thermal expansion** but too much -> noise & impact; **measured** by **fixing one gear** , **rocking the other** with a **dial indicator** at the **pitch circle** : (reading) = backlash. Or **feeler gauge** between non-driving flanks on assembly; **centre-distance** variation method.

4.5 Profile (involute) measurement

- **Involute profile** tester: **base disc (cylinder)** of **diameter = $D_b = D \cos \varphi$** mounted **co-axially with the gear** ; a **straight edge (ruler)** is held in contact with the base disc and **rolls without slip** ; a **stylus** touching the tooth flank moves by the **dial indicator - deviation from the true involute** = reading **(Hilger & Watts / Zeiss / Klingelnberg machines)** .

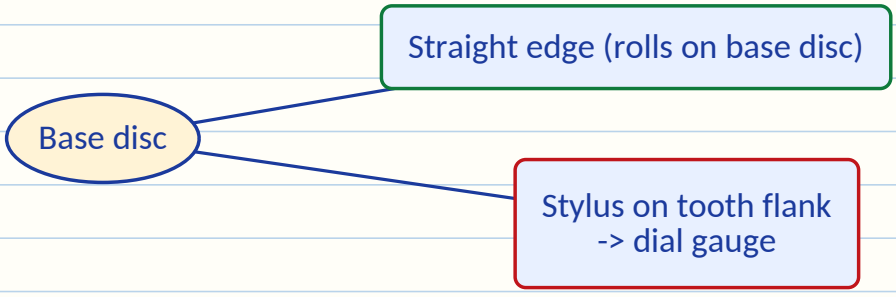


Fig 3.2: Principle of involute profile measurement (generating by rolling straight edge)

- **Optical profile projector (10x - 50x)** overlays the **magnified tooth profile** on a **drawn master chart** ; **toolmaker's microscope** ; **CMM (gear measurement**

software).

4.6 Pressure angle

- Gear pressure angle can be found from the base circle : $\cos \varphi = D_b / D$ or base pitch : $\cos \varphi = p_b / (\pi m)$. Measure base pitch with the base-pitch gauge and compare with the module (circular pitch). Also by profile projector with protractor reticle on the tooth flank at the pitch point.

4.7 Diameter of gear

- Outside (addendum) diameter : vernier caliper / micrometer (for even teeth directly; odd teeth need correction D_o = (reading) / cos (90/N)). Pitch circle diameter (PCD) : from measurement over rollers (pins) of diameter d placed in opposite tooth spaces : $M = D_b / \cos \varphi_1 + d$ (even teeth) - pin diameter (best size = 1.728 m for 20 degree); root diameter : micrometer across rollers at the root.

4.8 Tooth thickness - gear tooth vernier caliper

Tooth thickness is measured as chordal thickness (straight line between two points on the pitch circle), not arc thickness , with a gear tooth vernier caliper (two vernier scales : one horizontal for chord width w , vertical for the depth h (chordal addendum)).

Gear tooth vernier:

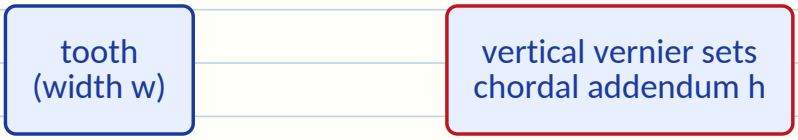


Fig 3.3: Gear-tooth vernier caliper

Chordal thickness and chordal addendum (spur gear, N teeth, module m)

Chordal thickness $w = N m \sin (90 / N)$

Chordal addendum $h = m + (N m / 2) [1 - \cos (90 / N)]$

- Arc tooth thickness = $\pi m / 2$ on pitch circle; chord is shorter. Vernier vertical scale set to h ; horizontal reads w - compare with computed.

Example : m = 4, N = 30: w = 30 x 4 x sin 3 degree = 120 x 0.05234 = 6.280 mm ; h = 4 + (120 / 2)(1 - cos 3 degree) = 4 + 60 x 0.001370 = 4.082 mm .

4.9 Constant chord method

- **Constant chord** = the **chord joining the points where the tooth flanks touch a standard rack (of the same module & pressure angle)** ; its length is **constant for all gears of the same module** and pressure angle - **independent of number of teeth** .

Constant chord formulae

$t_c = (\pi m / 2) \cos^2 \varphi$ (constant chord)

$h_c = m - (\pi m / 8) \sin 2\varphi$ (height of chord from tooth tip)

- **Example** m = 4, φ = 20 deg: t_c = 6.283 x 0.8830 = **5.548 mm** ; h_c = 4 - 1.5708 x 0.6428 = **2.990 mm** .

- **Advantages** : **no need to know the number of teeth** ; **no need to correct for each gear** ; suitable for **interchangeable gears** . Disadvantage: **depends on the vernier caliper depth setting accuracy** , needs **pressure angle** known.

4.10 Base tangent method

- **Span measurement** : measure the distance **W** across **k teeth** with a **disc micrometer / vernier (or flange) micrometer** whose anvils touch **opposite flanks** , **tangent to the base circle** (the common tangent = **base tangent length**). It is **independent of the gear axis** (no run-out effect) and **depends only on the base pitch** .

Base tangent length

$W = m \cos \varphi [\, \pi (k - 0.5) + N \, \text{inv}\, \varphi \,]$ with $\text{inv}\, \varphi = \tan \varphi - \varphi$ (rad) (inv 20 deg = 0.014904)

- Number of teeth to span: $k = N \varphi / 180 + 0.5$ (nearest integer); $W = (k - 1) p_b + s_b$ (base thickness).

Example : m = 4, N = 30, φ = 20 deg: k = 30 x 20 / 180 + 0.5 = 3.83 -> **4** ; W = 4 x 0.93969 [π (3.5) + 30 x 0.014904] = 3.75877 x (10.9956 + 0.4471) = **43.01 mm** .

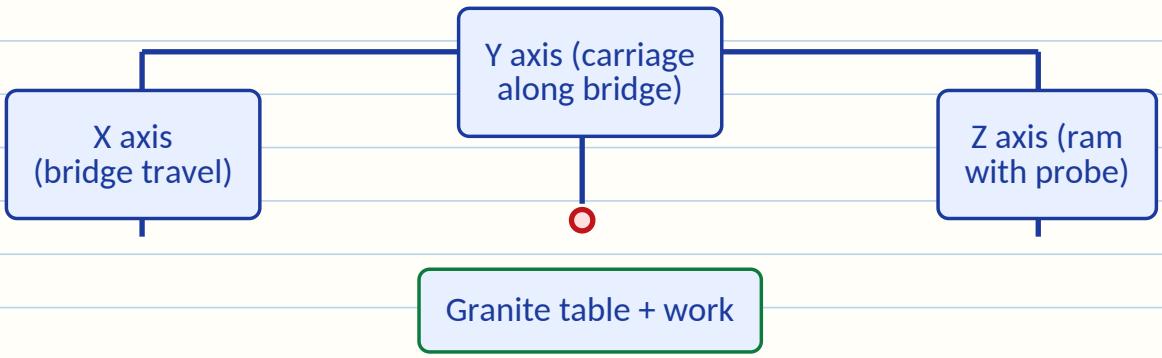
4.11 Composite error - Parkinson gear tester

- **Composite action test** : the gear under test is **meshed with a master gear** with **no backlash (double-flank contact)** under **spring pressure** ; one gear is on a **fixed slide** , the other on a **floating (sliding) carriage** ; **centre-distance variation** during **one revolution** is **recorded on a chart / dial** : **total composite error** = max variation; **tooth-to-tooth composite error** = max change per tooth. **Quick** and checks **overall quality** (run-out + profile + pitch).

5. Coordinate Measuring Machine (CMM)

A **Coordinate Measuring Machine (CMM)** is a **mechanical system** that moves a **measuring probe** to determine the **coordinates (X, Y, Z)** of points on a work-piece surface in a **3-D Cartesian system** . By measuring many points, **geometry** (dimensions, form, orientation, location) is computed by **software** . Called a **universal measuring machine** .

Bridge-type CMM (schematic)



Construction (main parts)

1. **Machine structure** : **bridge** , **cantilever** , **gantry** , **horizontal arm** , **column** types; made of **granite, aluminium, ceramic** - light, rigid, **low thermal expansion** ; **granite table (surface plate)** with T-slots / threaded inserts.
2. **Three mutually perpendicular axes X, Y, Z** guided on **air bearings** (frictionless, no wear); **drive** : motor / manual; **counterbalance** for Z axis.
3. **Measuring system (scales)** : **linear encoders / optical grating (Moire fringe), laser** on each axis - resolution **0.5 - 1 micron** ; some with **0.1 micron** .
4. **Probe system** : **touch-trigger probe** (kinematic seat with **three ball-rod contacts** ; the tip deflection breaks an electrical contact -> a **trigger signal** records the coordinates), **analog / scanning probe** (continuous displacement), **non-contact** (**laser, optical / video, white-light**), **probe head** (**indexable / motorized**), **stylus** (ruby ball tip, 0.3 - 8 mm) and **extensions** with **probe-changing rack** .
5. **Controller & computer** (**CNC**): motion control, data acquisition; **software** - alignment, geometry, **GD&T evaluation** , **CAD comparison** , **statistics** , reports.

Features / working

- **Measures** size, **form** (flatness, circularity, cylindricity), **position** , **orientation** , **profile** of **complex 3-D parts** in a **single set-up - versatile** ; **gears, cams, turbine**

blades, car bodies .

- Probe calibration with a reference sphere ; work coordinate system by alignment (plane-line-point, 3-2-1 method) - datum setting; auto-compensation for temperature and machine geometry errors (software error mapping).
- Operation modes : manual , motorized (joystick) , DCC (direct computer control; programmable CNC).
- Accuracy : typical volumetric accuracy = (1.5 + L/300) micron for L in mm; repeatability 0.5 micron .
- Advantages : flexibility , high accuracy , reduced set-up and inspection time , automatic data processing , no special gauges , operator independence , integration with CAD/CAM (reverse engineering).
- Limitations / error sources : cost , temperature & vibration , probe errors (lobing, pre-travel) , guideway geometry errors (21 errors for 3 axes), operator skill in programming , slow compared to gauging for mass production.

CMM type	Features / use
Bridge	most common , high accuracy, medium-large parts; fixed table
Cantilever	open access on three sides; small parts, good for sheet-metal ; less rigid
Gantry	very large parts (aircraft, car bodies); floor-mounted
Horizontal arm	car body inspection; large flat parts, dual-arm version
Portable arm / laser tracker	large assemblies, on-site; lower accuracy

Quick Revision / Important Questions

1. Elements of a screw thread; errors in threads. Two-wire & three-wire methods with formulae.
2. Floating carriage micrometer; thread gauges (plug, ring, snap); Taylor's principle for threads.
3. Gear tooth terminology; measurement of run-out, pitch, lead, backlash, profile.
4. Gear tooth vernier: chordal thickness; constant chord; base tangent method (examples).
5. CMM: construction, probes, features, advantages and limitations.

Unit 4

Measurement of Displacement and Strain

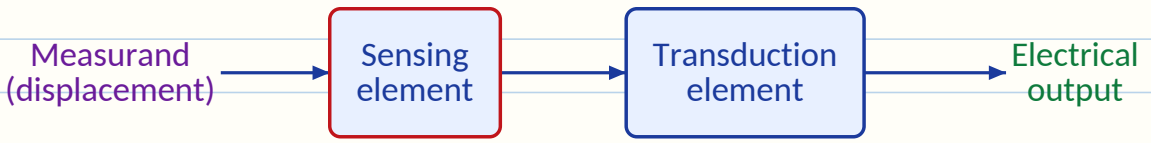
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Syllabus Topics Covered

1. Transducers: classification and requirements
2. Resistive (potentiometer) and inductive (LVDT) transducers
3. Capacitive transducers
4. Piezo-electric transducers
5. Ionization and photoelectric transducers
6. Calibration procedures
7. Electrical strain gauges and gauge factor
8. Use of strain gauges for bending, compression, tension and torque
9. Strain gauge rosettes

1. Transducers for Displacement Measurement

A **transducer** is a device that **converts one form of energy (the measured quantity) into another** - usually a **proportional electrical signal** (voltage, current, resistance, capacitance, inductance). **Displacement** (linear/angular position, thickness, level, strain, force, pressure, acceleration) is one of the commonly measured quantities.



Classification of transducers

Basis	Types	Examples

Active vs passive	**Active (self-generating)** : produce voltage/current without external power. **Passive** : need an **external excitation**	piezo-electric, thermocouple, photovoltaic / potentiometer, LVDT, strain gauge, capacitive
Primary vs secondary	primary senses (e.g. bellows -> displacement); secondary converts to electrical	diaphragm + LVDT
Analog vs digital	continuous output / pulse or coded output	LVDT / optical encoder
Principle	resistive, inductive, capacitive, piezo-electric, photoelectric, Hall effect, ionization	-

- **Requirements of a good transducer** : high **input-output linearity** , **sensitivity** , **repeatability** , **low hysteresis** , **good frequency response** , **small size / low weight** , minimum **loading effect** , **ruggedness** , **no sensitivity to environment (temperature, shock)** , **low cost** .

2. Resistive Transducers - Potentiometer

A **resistive potentiometer** (**pot**) has a **resistance element** (wire wound / conductive plastic / carbon film) with a **sliding contact (wiper)** attached to the moving object. The **output voltage** is a **fraction of the supply voltage** according to the **wiper position** .

Potentiometer

$e_o / e_i = x / x_t$ -> $e_o = e_i (x / x_t)$

- x = displacement of the wiper, x_t = total length of travel; e_i = excitation voltage (supply).
- Loading by a **load** R_m causes **non-linearity** ; the **loading error** is minimal when $R_m >> R_p$ (pot resistance).

- **Types** : **linear** (translatory, 1 mm to 500 mm), **rotary** (angular 0-360 degrees), **helical (multi-turn)** for **high resolution** . **Resolution** (for wire wound) = 1/N (N = no. of turns), about **0.1%** .
- **Advantages** : **simple, cheap, high output** , no amplifier required, both **AC & DC**

supply. **Disadvantages** : **wear** of contact (limited life), **friction** , **noise** , **low frequency response** , **resolution limited** , **loading effect** .

- **Uses** : **position feedback in servo mechanisms** , **machine tool slide position** , throttle/valve position, **level** (float) and **pressure** (Bourdon) indicators.

3. Inductive Transducers

Based on the **change in inductance** (self or mutual) of a coil due to the **movement** of a **ferromagnetic core / armature** , **change in air gap** or **coupling** . **Inductance L = N^2 / R_m** (N = turns, R_m = **reluctance** of the magnetic circuit).

3.1 Linear Variable Differential Transformer (LVDT)

The **most widely used inductive displacement transducer** : a **differential transformer** with **one primary and two identical secondary windings** , wound on a **hollow cylindrical former** ; a **movable soft-iron core** (ferromagnetic armature, moved by the **object** displacement) slides inside.

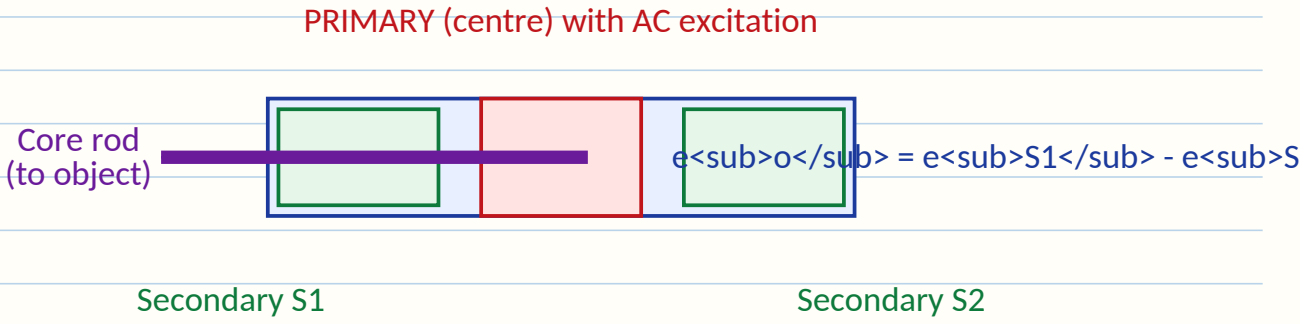


Fig 4.1: LVDT - construction

- **Working** : **AC (1-10 kHz, 3-15 V)** is applied to the primary. The **flux** links the secondaries through the **core** . **Secondaries are connected in series opposition** -> **net output e_o = e_S1 - e_S2** .

Core position	e_S1 vs e_S2	Output e_o
Central (null)	equal	**zero** (null position)
Displaced towards S1	e_S1 > e_S2	**in phase** with primary voltage; magnitude ∝ displacement
Displaced towards S2	e_S2 > e_S1	**180 degree out of phase** ; magnitude ∝ displacement

- **Phase-sensitive demodulator + filter** -> **DC output** with **polarity** indicating the **direction** . Output is **linear** over about **+-70% of the core range** .

- **Specifications** : range **+-0.1 mm to +-250 mm** ; **linearity 0.25%** ; sensitivity ~ 1-5 mV/V/mm; **resolution infinite** (practically 0.1 micron).
- **Advantages** : **no friction / wear** (no contact), **infinite resolution** , **high sensitivity** , **ruggedness** , **low hysteresis** , **repeatability** , **bidirectional** , **works in harsh environments** , **low power** consumption, **null reading is stable** . **Disadvantages** : need **AC excitation & demodulator** , **large core displacement** for small output, **sensitive to stray magnetic fields** , **dynamic response limited by carrier frequency (1/10)** .
- **Uses** : displacement (**gauging heads** , **comparators**), **thickness** , **pressure** (diaphragm), **force** (load cell), **acceleration** , **level** , **machine tool** position, **valve** position, **creep** and **roundness** .

3.2 Other inductive transducers

- **Variable reluctance** : **coil** + **moving armature** changes **air gap** -> inductance; **single (non-linear)** or **push-pull** (differential) types. Proximity sensing, **tachometer pick-up** (speed).
- **Mutual-inductance transducer** : two coils - mutual inductance changes with **separation / angle** (**synchros** , **resolvers** for angular position).
- **Eddy-current (proximity) transducer** : **high-frequency coil** - **eddy currents** in a nearby **conductive target** change the coil inductance / Q; measures **small distance (0.1 - 10 mm) non-contact** ; used for **shaft vibration** , **crack detection** , **thickness of coatings** .

4. Capacitive Transducers

Capacitance of a **parallel-plate capacitor** : **$C = \varepsilon_0 \varepsilon_r A / d$** ($\varepsilon_0$ = 8.854 pF/m; A = overlapping area; d = **gap** ; ε_r = dielectric constant). Displacement changes **d** , **A** or **ε_r** - hence **C** .



(a) change of gap d
C ∝ 1/d (non-linear)

(b) change of area A
C ∝ A (linear)

(c) change of dielectric
(level, thickness)

Principle	Relation	Remarks

Change of **gap** d	C = ε A / d; **ΔC / C = - Δd / d**	high sensitivity for **small displacements (um - mm)** ; **non-linear** -> use **differential** (push-pull) form
Change of **area** A	C ∝ A (linear)	**rotating vane** (angular) or **sliding plate** (linear); long range
Change of **dielectric** ε_r	slab of dielectric moves in/out	**liquid level** , **thickness / moisture** measurement

- **Measuring circuits** : **AC bridge** , **oscillator (frequency-modulated)** , **charge/voltage amplifier** , **operational amplifier** (output ∝ d for gap type: $e_o = - e_i (C_i / C_x) \propto d$).
- **Advantages** : **very small force needed** , **high sensitivity** , **excellent frequency response (MHz)** , **non-contact** , works at **high temperature** , **resolution < 0.01 micron** . **Disadvantages** : **stray capacitance** and **cable effects** , **temperature / humidity** , **non-linear** (gap type), needs **complex electronics** , **insulation** problems.
- **Uses** : **gap / thickness** (rolled sheets), **vibration** , **proximity** switches, **touch screens** , **liquid level** , **pressure (diaphragm)** , **micrometric positioning** (semiconductor lithography).

5. Piezo-electric Transducers

Piezo-electric effect : certain **crystals** (**quartz** , Rochelle salt, barium titanate, lead zirconate titanate **PZT**) develop an **electric charge** on their surface when **mechanically stressed** (force). The **reverse** effect: **deform** under an electric field (**ultrasonic generators**). Self-generating (**active**) transducers.

Charge and voltage

Q = d F (charge Q, force F, piezo-electric constant d (C/N): quartz **d = 2.3 pC/N**)

E_o = Q / C_p = d F / C_p , or **E_o = g t P** (g = voltage sensitivity (V m/N), t = thickness, P = pressure)

- C_p = capacitance of the crystal + cable.

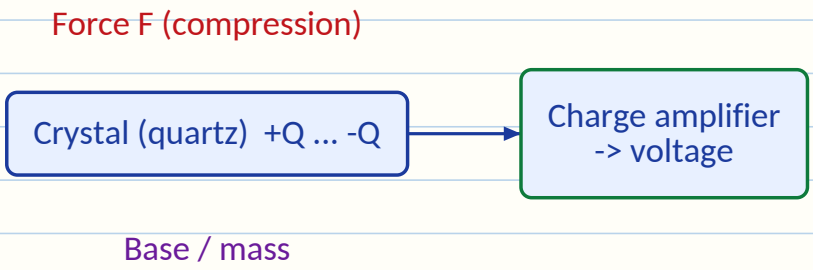


Fig 4.2: Piezo-electric transducer

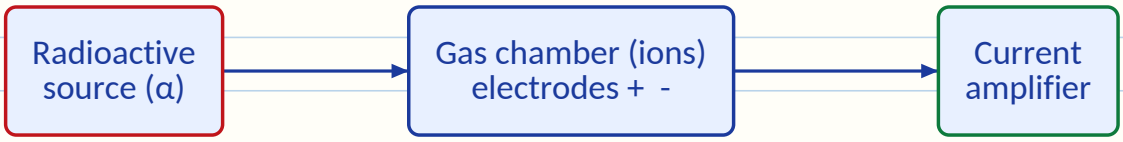
- **Measurement of displacement** : the force from a **spring-mass** (seismic) system or **diaphragm** is applied to the crystal; the **charge / voltage** is proportional to **force -> displacement, pressure, acceleration** . Practical for **dynamic** (rapidly changing) displacement / vibration.
- **Charge amplifier** (high input impedance): converts charge to voltage and **removes cable capacitance effect** ; **low-noise cable** needed.
- **Advantages** : **very high frequency response (up to 100 kHz+)** , **small, rugged** , **self-generating** , **high output** , wide range, **insensitive to temperature (quartz)** , **no external power** . **Disadvantages** : **can't measure static** quantities (charge **leaks** away: **only dynamic**), **high-impedance** electronics, **temperature effects (ceramics)** , **cable noise** , **pyro-electric** effects.
- **Uses** : **accelerometers** , **pressure** (engine knock, shock), **force / load cells** (dynamic), **vibration** , **ultrasonic transducers** , **igniters** , **microphones** , **acoustic emission** , **quartz crystal oscillator (clocks)** .

6. Ionization & Photoelectric Transducers

6.1 Ionization transducer

Ionization transducers use the **ionization of a gas** by a **radioactive source** (alpha particles) or by **an electric discharge** ; the **ion current** between two electrodes changes with the **geometry** (distance between electrodes, gas pressure).

- **Principle** : A **small radioactive source (Radium-226 / Americium-241 / Polonium)** ionizes the gas between **two electrodes** (**anode and cathode**) kept at a **potential** ; **ion current** (**saturation**) ∝ **number of ions collected** , which depends on the **path** and **gas pressure/density** . If one electrode **moves** (displacement), the **ion current changes** -> measure **displacement** with an **electrometer / amplifier** .



- **Other forms** : **ionization gauge (Bayard-Alpert)** for **low pressure (vacuum)** measurement; **flame ionization** & **smoke detectors** ; **radiation (beta gauge) thickness gauges** where absorption of rays through the **material** is measured.
- **Advantages** : **non-contact** , **high sensitivity** to small displacements, **no moving parts** , **very high-temperature** possible. **Disadvantages** : **radiation hazard** , **expensive** , **shielding** and **licence** , temperature/pressure dependence, **limited use** in practice.

6.2 Photoelectric transducers

Convert **light energy -> electrical signal** ; used for **non-contact** displacement, **position** , **counting** , **speed** and **dimension** measurement.

Type	Principle	Examples
Photo-emissive	light **ejects electrons** from a **cathode** in a vacuum tube -> **photo-current** ∝ intensity	phototube, **photomultiplier**
Photo-conductive	**resistance falls** when light falls on **semiconductor**	**LDR (CdS)** , photoconductive cell
Photo-voltaic	**generates EMF** when light falls on a **p-n junction**	**solar cell** , **photodiode** , **phototransistor**

- **Displacement measurement** : a **moving shutter / slit / mask** (attached to the object) **varies the amount of light** reaching a **photocell** -> output ∝ displacement (**linear**). **Reflective type** : **light reflected** from the object surface changes with the distance (**fibre-optic displacement sensor**).
- **Optical encoders** : **disc with transparent / opaque slots** (**incremental** pulses; **absolute** binary/gray code) with **LED + photodetector** -> **digital output** of angular / linear position (**resolution 1 / 10^4 turn** , **linear glass scales 0.5 - 5 micron**); also **laser (interferometer)** displacement.
- **Advantages** : **non-contact** , **fast response** , **high resolution** , **digital output** , **no wear** . **Disadvantages** : **ambient light** , **dust / oil contamination** , **temperature drift** , **aging** of cells, alignment of **optics** .

7. Calibration of Displacement Transducers

Calibration establishes the **relationship between the transducer output and the actual displacement** by comparing with a **standard** of higher accuracy.

1. Mount the transducer **rigidly** on a **calibrating stand** (**slip gauges** , **micrometer head** , **precision slide with vernier** or **laser interferometer** as reference).
2. **Warm up** the instrument; **adjust the zero** (null) and the **gain (span)** with the lowest & highest standard displacements.
3. Apply **known displacements** (e.g. 0, 10, 20, ..., 100% of range) in **steps upward (ascending)** and then **downward (descending)** , **recording the output** at each step (**3 to 5 cycles**).
4. **Plot** the **calibration curve** (output vs displacement): find **sensitivity (slope)** , **linearity** (best-fit straight line, % of full scale), **hysteresis** (difference up/down), **repeatability** , **zero drift** , **resolution** .
5. **Prepare a calibration chart / correction table** ; **certify** ; label the **next due date** .

- **Dynamic calibration** : apply a **step or sinusoidal** motion by a **vibration exciter (shaker)** to find **frequency response** and **time constants** .

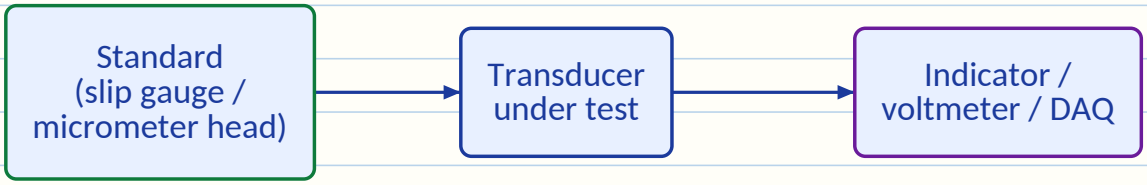


Fig 4.3: Static calibration set-up

8. Measurement of Strain

Strain (ε) = change in length / original length = ΔL / L (dimensionless; micro-strain, με = 10^{-6}). Under **Hooke's law σ = E ε** ; hence **stress can be found from strain** . **Electrical strain gauges** are the **most widely used** : they convert **strain into a change in resistance** .

Strain gauge principle & gauge factor

The **resistance of a conductor** R = ρ L / A. When the conductor is **stretched** , **length increases** , **area decreases** and **resistivity ρ changes** (piezo-resistive effect):

Gauge factor (strain sensitivity)

GF = (ΔR / R) / ε = 1 + 2ν + (Δρ/ρ)/ε

- ν = **Poisson's ratio** (0.3 for most metals) -> **1 + 2ν ≈ 1.6** ; the **piezo-resistive term** adds ~ 0.4 -> **GF ≈ 2 (metal foil: 1.9-2.1)** .
- **Semiconductor gauges (Si)** : **GF = 100 - 200** (up to +- 175; large piezo-resistive contribution).
- **ΔR = GF x R x ε** , e.g. R = 120 ohm, GF = 2, ε = 1000 με -> ΔR = 0.24 ohm.

Types of electrical strain gauges

Type	Construction	Features
Unbonded wire	wire stretched over insulated **pins** between a fixed frame and a movable armature	used in **transducers** (force, acceleration, pressure); **no cement** ; high stability
Bonded wire (grid)	**fine wire (0.025 mm)** in **grid** on **paper / plastic backing** , cemented to specimen	cheap; **earliest** type; poor heat dissipation
Metal foil	**etched foil grid** (2-5 micron) on **polyimide / epoxy backing** ; **most common**	**better heat dissipation** , **flexible shape** , **good fatigue life** , **stable**
Semiconductor	**silicon** strip with leads; **piezo-resistive**	**high GF 100-200** ; **non-linear** , temperature-sensitive, **brittle**
Weldable / capacitive / thin film	**spot welded** ; **capacitive** for high temp; **sputtered** film on diaphragm	**high-temperature** , **long term** field use

- **Materials** : **Constantan (Cu-Ni 45-55)** (GF 2.1, low temp coefficient - **most common**), **Isoelastic (Fe-Ni-Cr-Mo)** (GF 3.6, **dynamic**), **Nichrome V** (GF 2.1), **Karma** , **Platinum-tungsten** (GF 4, **high temp**), **Silicon** .
- **Gauge length** 0.2 - 100 mm (standard 6 mm); **resistance 120, 350, 1000 ohm** ; **backing** : paper, **polyimide** , **glass-fibre-reinforced epoxy** ; **adhesive** : cyanoacrylate, epoxy (**strain-transfer layer**).

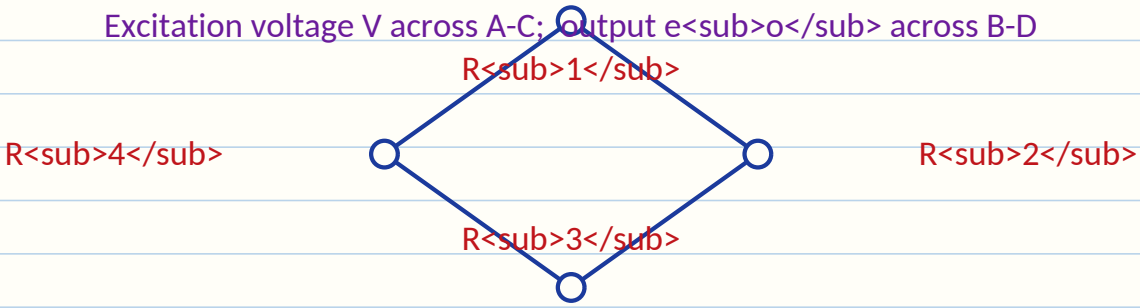
Installation (bonding) procedure

1. **Surface preparation** : degrease, abrade (emery 220-320 grit), **clean** with solvent (acetone), **mark alignment lines** , neutralize.
2. **Position** gauge on the marked axis, **apply adhesive** (thin), press with **thumb**

(gauge pressure) and **cure** (time/temperature).

3. **Solder leads** (**terminals**), **protective coating** (**moisture-proof** : wax, silicone), **check resistance & insulation** (> 1000 M-ohm).

Measuring circuit: Wheatstone bridge



Bridge output

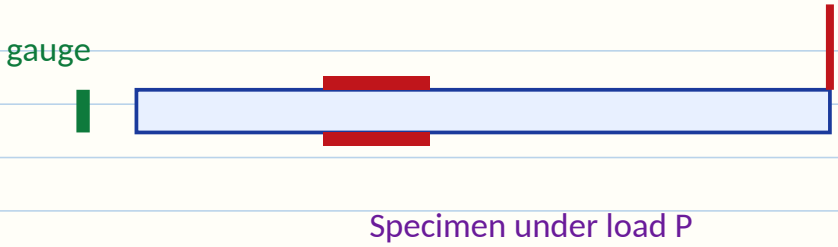
$e_o = V [R_1 / (R_1 + R_2) - R_4 / (R_3 + R_4)]$; balanced when $R_1R_3 = R_2R_4$.

- **Quarter bridge** (one active gauge): $e_o = V (\Delta R/R)/4 = V\ GF\ \varepsilon / 4$.
- **Half bridge** (2 active gauges): $e_o = V\ GF\ \varepsilon / 2$ (e.g. bending).
- **Full bridge** (4 active): $e_o = V\ GF\ \varepsilon$ - **largest output, temperature compensated** .

- **Temperature compensation** : a **dummy gauge** (same material & type, unstrained, on the same material) in the **adjacent arm** of the bridge cancels the resistance change due to temperature; **self-temperature-compensated (STC)** gauges are also available.
- **Example** : V = 10 V, GF = 2, ε = 500 με, quarter bridge -> $e_o = 10 \times 2 \times 500 \times 10^{-6} / 4$ = **2.5 mV** ; **full bridge = 10 mV** .

9. Use of Strain Gauges - Bending, Tension, Compression, Torque

9.1 Tension and compression (axial loading)



Specimen under load P

- **Single active gauge** on the member along the load axis: $\varepsilon = \sigma / E = P / (A E)$;

$\Delta R/R = GF\ \varepsilon$; **load $P = \varepsilon A E$** .

- **Poisson gauge** (**second gauge transverse** , inactive for axial strain but $\varepsilon_t = -\nu\ \varepsilon$) in an **adjacent arm** -> **output increases by (1 + ν)** : $e_o = V\ GF\ \varepsilon\ (1 + \nu) / 4$. **Dummy gauge** gives **temperature compensation** .
- **Four-gauge full bridge** on a **column / tension bar** : two axial (R_1, R_3, opposite arms) + two transverse (R_2, R_4): **output = V GF ε (1 + ν) / 2** ; **eliminates bending effects & temperature** . Used for **load cells** .

9.2 Bending (cantilever / beam)

- In a **cantilever** beam loaded at the free end, the **top fibre is in tension (+ε)** and the **bottom fibre in compression (-ε)** with equal magnitude. Mount **gauge R_1 on top and R_2 on bottom** , in **adjacent arms** of the bridge -> **output doubles** (half bridge: $e_o = V\ GF\ \varepsilon / 2$), **temperature compensated automatically** , **axial load cancelled** .
- **Strain at gauge : $\varepsilon = 6 P L / (E b t^2)$** (rectangular section, L = distance from the load to the gauge, b = width, t = thickness); **bending moment M = P L** ; **stress $\sigma = 6M/(b t^2) = E \varepsilon$** .
- **Full bridge for bending : two gauges on top (R_1, R_3) + two on bottom (R_2, R_4)** -> output **= V GF ε** (4 times the quarter-bridge). Used in **bending-beam load cells** , **weighing scales** .

9.3 Torque measurement (shaft)

A shaft in **pure torsion** has **principal strains at +45 degrees (tension) and -45 degrees (compression)** to its axis; **shear strain γ** along the axes. **Gauges are bonded at 45 degrees** to the shaft axis.

Shaft with 4 gauges at +-45 degrees (full bridge)



Torque from strain

Shear stress $\tau = G\gamma$; $\varepsilon_{45} = \gamma / 2 = \tau / (2G)$ (magnitude equal at +45 and -45; opposite sign).

$\tau = G\gamma = 2G\varepsilon_{45}$; solid shaft: $T = \pi d^3 \tau / 16 = \pi d^3 G \varepsilon_{45} / 8$, and with $G = E / 2(1 + \nu)$: $T = \pi d^3 E \varepsilon_{45} / (16(1 + \nu))$.

- **Full bridge (4 gauges, +45, -45, +45, -45)** : $e_o = V\,GF\,\varepsilon_{45}$ - four times output, **cancels bending & axial loads** and temperature. **Slip rings / telemetry** needed for rotating shafts.

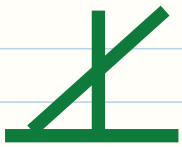
- **Example** : solid steel shaft d = 40 mm, E = 200 GPa, ν = 0.3, measured ε_{45} = 500 με -> $T = \pi \times 0.04^3 \times 200e9 \times 500e\text{-}6 / (16 \times 1.3) = 966$ N.m (approx.).

Practical points

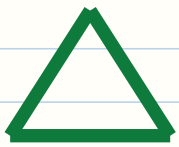
- **Use of dummy gauges** & **bridge balancing** ; **lead-wire resistance** compensation (**three-wire** connection); **shunt calibration** (a known resistor across one arm simulates a known strain); **zero drift** , **self-heating** (limit gauge current < 25-30 mA), **hysteresis** , **creep** , **cross sensitivity** (transverse strain).

10. Strain Gauge Rosettes

When the **directions of principal strains are unknown** (a biaxial stress field), strain at a point must be measured in **at least three directions** . A **rosette** = **three (or more) strain gauges arranged at fixed angles** on a **common backing** .



Rectangular rosette
0 / 45 / 90 degree



Delta rosette
0 / 60 / 120 degree



Tee rosette (two gauges at 90 degree)

Rectangular (0 - 45 - 90) rosette: gauges a, b, c with strains ε_a, ε_b, ε_c

Principal strains : $\varepsilon_1,2 = (\varepsilon_a + \varepsilon_c) / 2 \pm (1/\sqrt{2})\ \mathrm{sqrt}[\ (\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2\ ]$

Direction : $\tan 2\theta = (2\varepsilon_b - \varepsilon_a - \varepsilon_c) / (\varepsilon_a - \varepsilon_c)$ (θ from gauge a to ε_1)

Principal stresses : $\sigma_1 = E(\varepsilon_1 + \nu\varepsilon_2) / (1 - \nu^2)$, $\sigma_2 = E(\varepsilon_2 + \nu\varepsilon_1) / (1 - \nu^2)$

Delta (0 - 60 - 120) rosette

$\varepsilon_{1,2} = (\varepsilon_a + \varepsilon_b + \varepsilon_c) / 3 \pm (\sqrt{2} / 3) \sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2 + (\varepsilon_c - \varepsilon_a)^2]}$

Example (rectangular rosette) : $\varepsilon_a = 800$, $\varepsilon_b = 400$, $\varepsilon_c = -200$ με -> $(\varepsilon_a + \varepsilon_c)/2 = 300$; $\sqrt{[(400)^2 + (600)^2]} = 721.1$; $/\sqrt{2} = 509.9$ -> **$\varepsilon_1 = 810$ με, $\varepsilon_2 = -210$ με** ; $\tan 2\theta = (800 - 800 + 200)/(1000) = 0.2$ -> **$\theta = 5.7$ degree** .

- **Stacked / planar rosettes** : **stacked** (gauges one above another) for **small spots** ; **planar** (side by side) for **cooler operation** . **Uses** : **stress analysis** of **pressure vessels, aircraft structure, bridges** , **welded joints** , **residual stress** (hole drilling).
- **Selection** : **rectangular** for **known principal directions** or **simple** reduction; **delta** has **equal angle** (uniform), **better** when **no preference** about direction.

Quick Revision / Important Questions

1. LVDT: construction, working, characteristics, advantages. Potentiometer and loading effect.
2. Capacitive and piezo-electric transducers; charge amplifier.
3. Ionization and photoelectric transducers; calibration procedure.
4. Strain gauge: gauge factor derivation, types, Wheatstone bridge, temperature compensation.
5. Strain gauges for bending, tension / compression and torque; rosettes (examples).

Unit 5

Measurement of Force, Torque and Pressure

(Handwritten Notes)

Syllabus Topics Covered

1. Force: direct methods - analytical balance, platform balance
2. Elastic members: load cells, cantilever beams, proving rings
3. Torque: torsion bar, servo-controlled and absorption dynamometers
4. Pressure: standards and calibration, basic methods
5. Dead weight gauges and manometers
6. High and low pressure measurement
7. Elastic pressure transducers

1. Measurement of Force

Force ($F = m\,a$, unit **newton N**) is a **vector** quantity that **changes the motion or shape** of a body. Methods of measuring force fall into two classes:

1. **Direct (balancing) methods** - **compare** the unknown force with a **known gravitational force** (known mass) or with a **magnitude** produced by a lever, for example **analytical balance, platform balance** .
2. **Indirect methods** - **measure the effect** of the force: **deflection of an elastic member** (spring) -> **load cell, proving ring** , **hydraulic / pneumatic** pressure balancing, **magnetic / piezo-electric** effects, **acceleration** of a known mass.

1.1 Direct method: Analytical balance

A **very precise equal-arm balance** used to measure **small masses** (0.1 mg precision) -> **weight = force** . Works on the **principle of the moment (lever)** : at balance, $W_1 \times L_1 = W_2 \times L_2$.

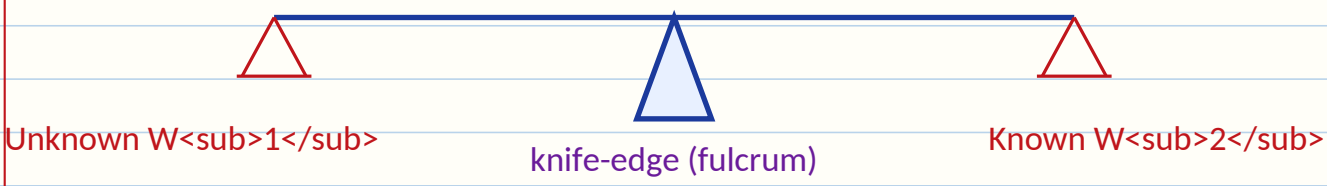


Fig 5.1: Equal-arm (analytical) balance

- **Construction** : **beam** (light, rigid, with **three agate knife edges** - central fulcrum, two outer for pans), **pointer** moving on a **scale** (sensitivity), **pans** , **beam arrest** , **levelling screws** , **glass case** (draught protection), **rider / chain** for fractional weights.
- **Equal arms** -> $W_1 = W_2$ independent of the arm length (if equal) - **double-weighing (Gauss)** method eliminates **arm-length inequality** .
- **Sensitivity** : S = **angular deflection per unit mass** = **(L) / (W_b h)** : increases with **longer arms** , **lighter beam** , and **CG nearer the fulcrum** (small h).
- **Modern** : **electronic (electromagnetic force-restoration)** balances; **resolution 1 microgram** .

1.2 Platform (weighing) balance

Used to weigh **heavy loads** (up to tonnes): the **load on a platform** is transmitted through a **system of levers** to a **beam** balanced by **small known weights** (poise), the ratio of lever arms (e.g. **1 : 10 : 100**) reduces the force.

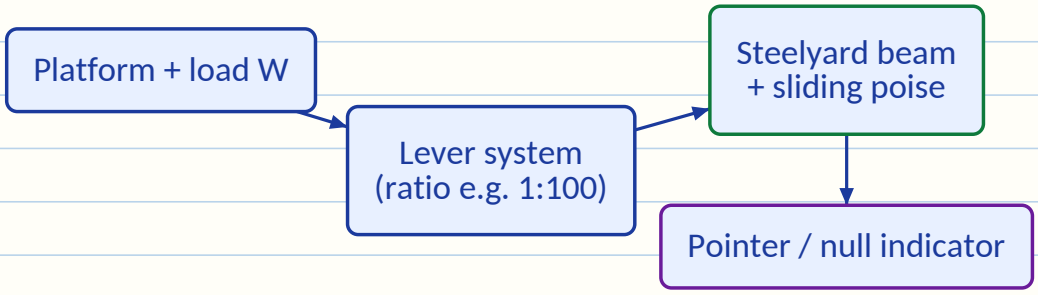


Fig 5.2: Platform balance (block diagram)

- **Principle** : load W acts on platform -> **main lever** -> **steelyard rod** -> **beam** ; the **poise** weight w is slid until the beam is **horizontal (null)** -> **W = w x lever ratio** .
- **Advantages of direct (balance) methods** : **high accuracy** , **independent of elasticity** , **simple** , **no calibration drift** . **Limitations** : **slow** , **not suitable for dynamic** force, **location-dependent g** (gravity) effect, **manual** .

2. Elastic Force Measuring Members

Force is applied to an **elastic member** (of known **spring constant**) and the **resulting deformation (strain / deflection)** is measured by a **displacement transducer / strain gauge** . The member's behaviour obeys **Hooke's law : F = k x** (within elastic limit).

1. **Spring balance** (helical): extension ∝ load; **simple, low accuracy** .
2. **Load cells** (**strain-gauge, hydraulic, pneumatic, piezo, magnetoelastic**).
3. **Cantilever beam** ; **proving ring** ; **elastic ring** ; **column** .

2.1 Load cells

A **load cell** is a transducer that converts **force (load) into an electrical signal** . The **elastic body** carries **strain gauges** wired in a **Wheatstone bridge** ; **output voltage ∝ force** .

Load cell type	Elastic element	Features
Column (compression/tension)	solid / hollow **steel cylinder** with 4 gauges: 2 axial, 2 transverse (Poisson)	high capacity (kN - MN); **very rigid** ; compensates bending
Cantilever (bending beam)	**beam** fixed at one end; load at free end; gauges **top & bottom**	**low capacity (up to ~ 5 kN)** ; high output; **weighing scales**
Shear beam	**web with shear strain** at 45 degrees	**insensitive to load position** ; platform scales
Ring / S-type	ring or S-shaped member	tension & compression; **medium capacity**
Hydraulic	**piston + diaphragm** with **oil** ; **pressure** ∝ force (P = F/A) read by a **gauge**	**no electric power** ; **hazardous areas** ; **very high capacity**
Pneumatic	diaphragm and **air back-pressure** (nozzle-flapper)	**clean, safe** ; **medium accuracy** ; food / chemical plants

Strain-gauge load cell

Load cell output = (V_exc x GF x ε) for full bridge; ε = F / (A E) (column); sensitivity expressed in mV/V (e.g. 2 mV/V at rated capacity).

- Accuracy 0.02 - 0.1% of rated load; temperature compensated .

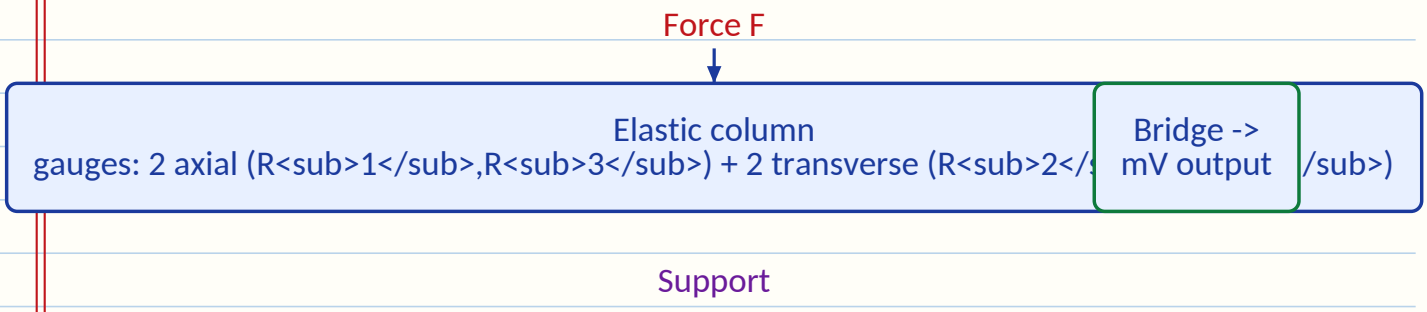


Fig 5.3: Strain-gauge load cell

- Advantages : wide range (g to MN) , compact , good accuracy , fast response , remote reading , used in industrial weighing , material testing machines , crane overload protection . Disadvantages : temperature/creep , overload damage , need for excitation, amplifier , side-load sensitivity .

2.2 Cantilever beam

A simple beam rigidly fixed at one end with the force applied at the free end . The deflection (or strain at the surface) is proportional to the force.

Cantilever relations

Deflection at free end: $\delta = F L^3 / (3 E I)$, with $I = b t^3 / 12$ (rectangular).

Surface strain near the root: $\varepsilon = 6 F L_g / (E b t^2)$ (L_g = distance from the load to gauge)

-> $F = \varepsilon E b t^2 / (6 L_g)$: top gauge in tension, bottom gauge in compression -> half/full bridge, doubled output, temperature-compensated.

- Example : steel beam b = 20 mm, t = 3 mm, L = 150 mm, E = 200 GPa, measured ε = 500 με: F = 500e-6 x 200e9 x 0.02 x $(0.003)^2$ / (6 x 0.15) = 20 N .
- Uses : low forces (weighing, dynamometer for machining forces), accelerometers (mass at the tip), micro-force sensors , AFM probes .

2.3 Proving ring

A proving ring is a steel ring (annular) elastic member with two diametrically opposite loading points ; the change in diameter (deflection) under load is measured with a micrometer / dial gauge / LVDT and calibrated to give the force .

Used as a **secondary standard** for **calibrating tensile & compression testing machines** (**load calibrations**).

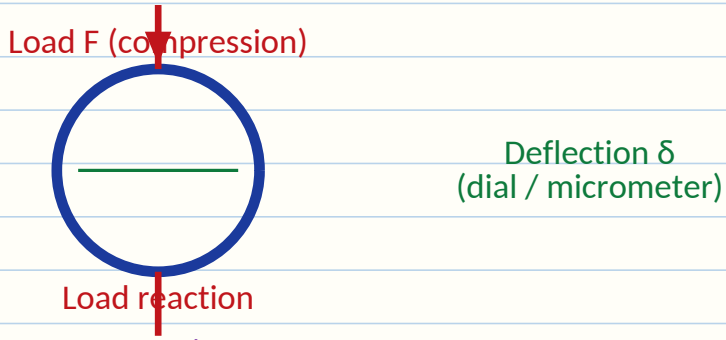


Fig 5.4: Proving ring

Deflection of proving ring

$\delta = 0.149\ F\ R^3 / (E\ I)$ (R = mean radius; $I = b\ t^3 / 12$; b = axial width, t = wall thickness)

- Linear response; **sensitivity increases with R** , decreases with t.

- **Features** : **simple, rugged** , **high accuracy (+-0.1%)** , **can be used in compression or tension** ; **vibrating-reed** between the diametral points (micrometer + reed resonance indicator) for **precise determination** of the diameter change; **temperature** corrections needed (**steel E changes**).
- **Capacity** : 2 kN - 2 MN; **calibration standard** as per **IS 1828 / ISO 376** .

Force measuring device	Principle	Typical use
Analytical / platform balance	direct, lever	mass / weights
Load cell	strain gauge on elastic body	industrial weighing, testing
Cantilever beam	bending strain/deflection	small forces
Proving ring	ring deflection	calibration of testing machines

3. Measurement of Torque

Torque (T = F x r) is the **twisting moment** acting on a shaft (N.m). It is measured by a **dynamometer** , and **power** is found from torque and speed:

Power transmitted

$P = T\omega = 2\pi N T / 60$ (watts, with T in N.m and N in rpm)

$kW = 2\pi N T / (60 \times 1000)$.

Class of dynamometer	Action	Examples
Absorption	absorb the power developed by the prime mover - **convert to heat** or electricity	**Prony brake** , **rope brake** , **hydraulic (water brake)** , **eddy-current** , **electric generator**
Driving	**drive** the machine and measure the **input** torque (**loading machines**)	electric motor dynamometer
Transmission	**transmit** power from engine to load, **measuring the torque in between**	**torsion bar** , belt, **epicyclic gear** , strain-gauge shaft

3.1 Torsion bar (transmission) dynamometer

The **power is transmitted through a shaft (torsion bar)** of **known torsional rigidity (G J)** ; the **angle of twist** θ between the two ends is **proportional to torque** .

Torsion

$T / J = G\theta / L = \tau / r \rightarrow T = G J \theta / L$ ($J = \pi d^4 / 32$ for a solid shaft; G = modulus of rigidity; L = gauge length).

- **Twist angle is small** - amplified by **optical lever** , **mechanical (dial) gauge** , **strain gauges (+-45 degree)** , **magnetic pickups** , **phase shift between two toothed discs (inductive)** , **slotted discs (stroboscopic / photoelectric)** .

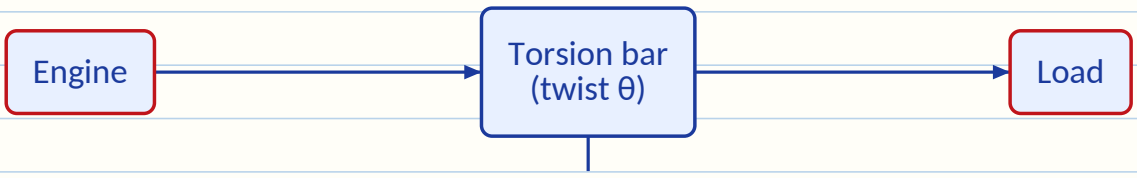


Fig 5.5: Torsion-bar dynamometer

- **Phase-difference (magnetic) type** : **two toothed wheels** at the ends of the bar with **magnetic pickups** : at no load, the pulses are **in phase** ; at torque the **time (phase) difference** $\Delta t \propto$ twist: $\theta = \omega \Delta t \rightarrow T = G J \omega \Delta t / L$; gives **torque & speed** on **rotating shaft** (contact-less, **no slip rings**).
- **Strain-gauge torque shaft** : **four gauges at 45 degrees** on shaft in **full bridge** ; signal by **slip rings / rotary transformer / telemetry** .
- **Advantages** : **measures running torque without absorbing the power** , **simple** , **good accuracy (0.25-1%)** . **Disadvantages** : **bar flexibility** may affect the system dynamics, **limited by shaft strength** , **signal transfer** problems.

3.2 Servo-controlled (electrical) dynamometer

A **cradled (trunnion-mounted) electrical machine** (DC / AC) whose **stator is free to rotate** a small angle: the **reaction torque on the stator = the torque on the shaft** . A **null-balance servo** automatically **restores** the stator position.

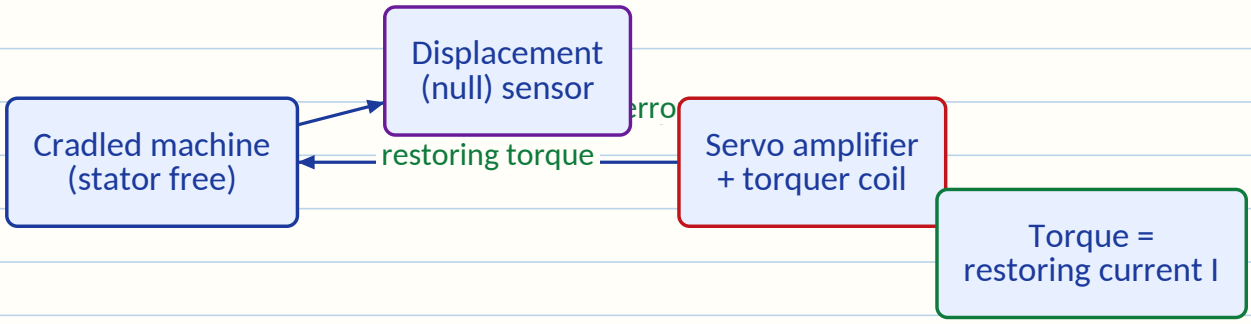


Fig 5.6: Servo-controlled dynamometer

- **Working** : when torque is applied, the **stator tends to rotate** ; the **sensor (LVDT, optical, capacitive)** detects the **displacement** ; the **servo amplifier** passes **current through a torque motor / coil** that **opposes** the movement till the **stator returns to null** ; **current I ∝ torque (T = K I)** ; read as **torque (N.m)** by a calibrated **ammeter** .
- **Advantages** : **no moving mechanical parts of measurement** - **high precision** , **fast** response, **continuous / automatic** , **can both absorb & drive (4-quadrant)** , works at **all speeds** , **remote and computer control** . **Disadvantages** : **costly** , **complex electronics** ; **limited power** range; **bearing friction** and **windage** corrections.

3.3 Absorption dynamometers

(a) Prony brake

- A **mechanical friction brake** : a **flywheel** (brake drum) rotating inside **wooden blocks** clamped by **bolts** , with a **lever arm** carrying a **dead weight W** (or a

spring balance) at a **distance L** from the axis; the **friction torque = W x L** (balance).

Prony brake

T = W L ; **Brake power BP = 2π N W L / 60** (W in N, L in m).

- Heat produced is removed by **water cooling** of the drum.

- **Disadvantages** : **unsteady** (coefficient of friction changes with **heat** , **self-energising** - jerks), **only for low power / speed** , **needs cooling** . **Advantages** : cheap, simple.

(b) Rope brake dynamometer

- A **rope (several turns)** wound round the **flywheel (drum)** ; one end carries **dead weight W** , the other a **spring balance S** ; the **difference** (W - S) is the **friction force** : **T = (W - S) (D + d) / 2** , where **D = drum diameter, d = rope diameter** .
- **BP = π N (W - S)(D + d) / 60** . Used for **small engines** ; **cooling water** in the drum; **rope wear** ; **unsteady** for large powers.

(c) Hydraulic (water-brake) dynamometer (Froude)

- A **rotor (disc with pockets)** rotates in a **casing (stator)** filled with **water** ; the **shear/vortex (churning) of water** absorbs power and **heats the water** ; the casing is **trunnion mounted** and the **reaction torque** is read through a **torque arm and a spring/dead weight** (**T = W L**). **Absorption controlled by water level / sluice plates** .
- **Advantages** : **high power & speed (MW)** , **compact** , **stable** , **continuous load control** . **Disadvantages** : **water supply / disposal** , **corrosion** , **cannot be run at low speed** , **braking torque ∝ N^2** .

(d) Eddy-current & electric dynamometers

- **Eddy-current** : a **toothed (or plain) rotor disc** turning in the **magnetic field of an excited stator coil** -> **eddy currents** in the **stator / rotor** produce **drag torque** , **power dissipated as heat** (water-cooled). **Torque ∝ excitation current** ; fast **response** , **no mechanical wear** , widely used for **engine testing** . **DC / AC generator type (cradled)** - **regenerative** , **absorbs or drives** .
- **Fan brake** , **air brake** : dissipate power by **fan** , for small / **high speed** machines (turbo).

4. Measurement of Pressure

Pressure = force per unit area ($P = F / A$) ; SI unit pascal ($Pa = N/m^2$) . Types:

- Absolute pressure - above perfect vacuum (zero reference). Gauge pressure - above atmospheric : P_abs = P_gauge + P_{atm} .
- Vacuum pressure - below atmospheric : P_abs = P_{atm} - P_vac. Differential pressure = difference between two pressures (P_1 - P_2).

Unit	Equivalent
1 bar	10^5 Pa = 100 kPa
1 atm (standard)	101.325 kPa = 760 mm Hg = 1.01325 bar
1 mm Hg (torr)	133.322 Pa
1 kgf/cm^2	98.07 kPa
1 psi	6.895 kPa

Standards and calibration

- Primary standards : dead-weight tester (piston gauge) and liquid-column manometers (mercury) - pressure derived from fundamental quantities (mass, length, g) ; accuracy 0.005 - 0.02% .
- Secondary standards : precision elastic gauges (test gauges, Bourdon / quartz), calibrated against primary ; working standards : industrial gauges and transducers .

1. Calibration procedure : connect the gauge under test to the dead-weight tester (hydraulic / pneumatic); apply known loads in steps upward (0, 20, 40 ... 100% of range) and downward ; record the gauge readings ; compute errors, hysteresis ; tap gently to overcome friction; apply corrections for local g , air buoyancy , temperature , head of oil (height difference); issue a calibration certificate .

Basic methods of pressure measurement

1. Balance (gravitational) methods : liquid column (manometer) , dead weight (piston) gauge .
2. Elastic methods : deformation of an elastic member - Bourdon tube, diaphragm, bellows, capsule .
3. Electrical methods : strain gauge, capacitive, piezo-electric, LVDT, potentiometric , resonant (vibrating wire) transducers.
4. Physical property methods (for vacuum): thermal conductivity (Pirani,

thermocouple) , ionization , viscosity .

5. Dead-Weight Gauges and Manometers

5.1 Dead-weight gauge (piston gauge tester)

Pressure is generated in a fluid (oil) by a piston-cylinder assembly of precisely known area A loaded with known weights W ; the pressure balanced is P = W / A (including the piston and carrier weight). Used to calibrate pressure gauges (up to 1000+ MPa).

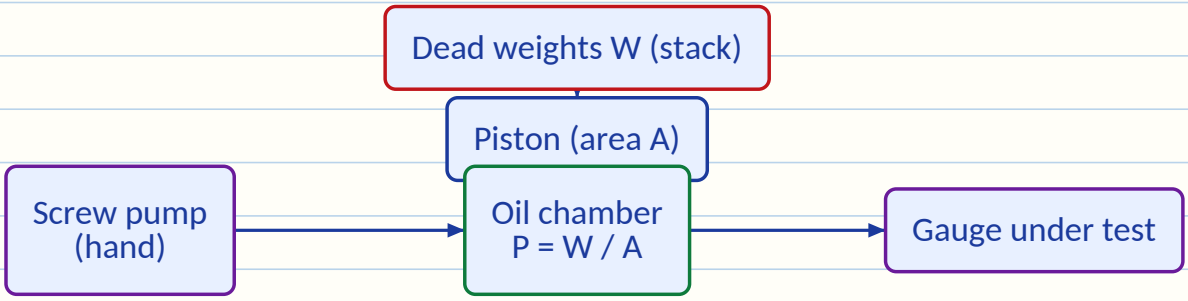


Fig 5.7: Dead-weight tester

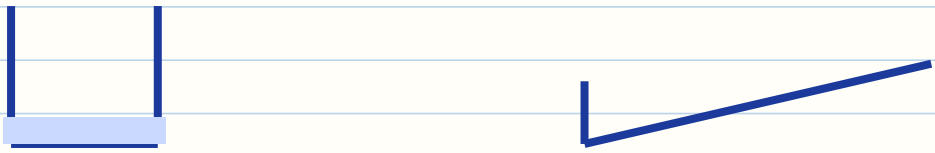
- Working : screw pump raises the oil pressure until the loaded piston floats (lifts slightly); the piston is rotated (to reduce static friction); then P = W / A equals the gauge pressure; the gauge under test is compared. Effective area A = (A_p + A_c) / 2 (piston and cylinder) and depends on pressure & temperature .
- Accuracy +-0.02% ; uses : calibration labs , standard for pressure transducers . Pneumatic version for low pressure (gas lubricated piston).
- Corrections : local g , air buoyancy , temperature of piston , height difference between piston reference level and gauge : ΔP = ρ g h .

5.2 Manometers (liquid-column gauges)

Measure pressure by balancing it with the height of a liquid column (Mercury, water, alcohol or coloured oil): P = ρ g h .

Manometer	Principle / equation	Notes
U-tube	$P_1 - P_2 = \rho g h$ (h = difference in levels)	simple , range up to ~ 2 bar (Hg), both gauge & differential; zero setting
Well (cistern) type	$P = \rho g h (1 + A_t / A_w)$; with large well area, single reading only on the tube	easier reading; barometer (Fortin)

Inclined tube	P = ρ g L sin θ : h magnified by 1 / sin θ	small pressures (draft gauges) ; sensitivity increases as θ decreases (typically 10 degree)
Micro-manometer	optical / micrometer reading of the level	very small pressure differences (0.01 mm of water)



(a) U-tube (b) Inclined tube

- Advantages : simple, cheap, accurate , no calibration needed (primary) .
  Disadvantages : bulky, fragile , not for high pressure or dynamic , temperature and capillarity effects (correction $\Delta h = 4\sigma \cos\theta / (\rho g d)$ - use tubes > 12 mm), meniscus reading, mercury toxicity .

6. High and Low Pressure Measurement

6.1 High-pressure measurement

For pressures above ~ 1000 bar (100 MPa) , Bourdon gauges are replaced by electrical (resistance) methods.

- Bridgman gauge (manganin wire gauge): a coil of manganin wire (alloy of 84% Cu, 12% Mn, 4% Ni) is subjected to the fluid pressure inside a pressure vessel ; its resistance changes linearly with pressure ($\Delta R / R = K P$, $K \approx 2.3 \times 10^{-6}$ per bar , i.e. pressure coefficient ≈ 2.5×10^{-11} per Pa); read by a Wheatstone bridge ; range up to 1 - 10 GPa .

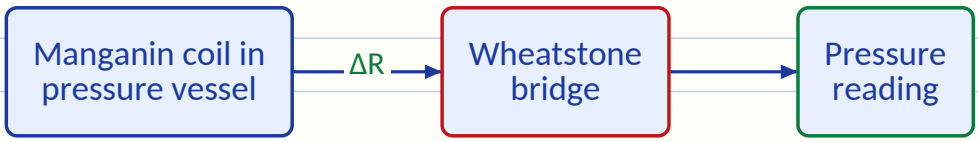


Fig 5.8: Bridgman gauge

- Features of manganin : high & stable pressure coefficient , very low temperature coefficient of resistance; hysteresis small ; calibrated against a dead weight tester . Used for explosive / shock, hydraulics & materials research .
- Other : strain-gauge (thick-diaphragm) transducers , heavy-walled Bourdon tubes , piezo-electric (dynamic), Piston gauge (to 1 GPa) .

6.2 Low-pressure (vacuum) measurement

Vacuum ranges: rough (1 - 760 torr) , medium (10^{-3} - 1 torr) , high (10^{-6} - 10^{-3}) , ultra-high (< 10^{-9} torr) . Units: torr, micron (1 micron = 10^{-3} torr) .

Gauge	Principle	Range (torr)
McLeod gauge	Boyle's law (compress a sample)	10^{-6} - 10
Pirani gauge	thermal conductivity (resistance of hot wire)	10^{-4} - 1
Thermocouple gauge	temperature of heated filament by thermocouple	10^{-3} - 1
Ionization gauge (hot cathode / Bayard-Alpert)	ion current ∝ gas density	10^{-10} - 10^{-3}
Knudsen / viscosity gauge	molecular momentum transfer	10^{-7} - 10^{-3}
Capacitance diaphragm	diaphragm deflection with capacitance	10^{-4} - 10^{3}

McLeod gauge

A primary standard for vacuum: it compresses a known large volume V of low-pressure gas into a small capillary volume (by raising mercury), then the pressure of the compressed gas is determined from the height h of mercury; Boyle's law gives the original pressure .

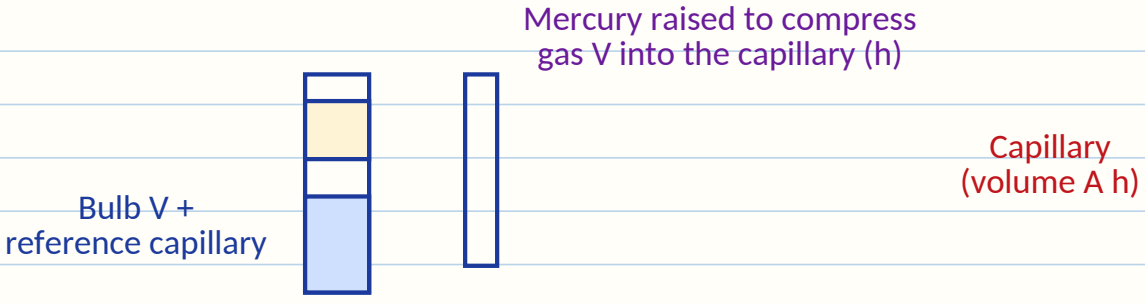


Fig 5.9: McLeod gauge

McLeod gauge equation

$P_1 V = P_2 (A h)$ with $P_2 = P_1 + h$ (mm Hg)

$P_1 = A h^2 / (V - A h) \approx A h^2 / V$ (square-law scale)

- A = capillary cross-section area, h = mercury height difference , V = volume of bulb + capillary.

- Advantages : absolute (independent of gas type) for permanent gases , calibration standard . Disadvantages : intermittent , not for condensable vapours (water vapour), mercury hazards, no continuous reading .

Pirani gauge

- A heated filament (tungsten / platinum) in the vacuum; heat loss by gas conduction depends on pressure (thermal conductivity ∝ P at low pressures). The filament resistance (temperature) changes and is measured by Wheatstone bridge (constant-temperature mode). Continuous , fast , range 10^{-4} - 1 torr ; gas-dependent calibration .

Ionization gauge

- Hot-cathode : electrons emitted by a heated filament ionize the gas molecules; ions collected at a negative collector give a current $I_c = S I_e P$ (S = sensitivity, I_e = emission current). Bayard-Alpert design with thin wire collector (reduces X-ray effect) reads down to 10^{-10} torr ; cold-cathode (Penning) type uses magnetic field and discharge ; fragile filament , gas-dependent .

7. Elastic Pressure Transducers

Pressure acts on an elastic member , which deforms (deflection ∝ pressure); the deflection is indicated mechanically (pointer) or sensed electrically (strain gauge, LVDT, capacitive, potentiometer).

7.1 Bourdon tube

A curved, flattened (elliptical cross-section) metal tube (C-shape , spiral , helical or twisted) closed at one end ; the open end is fixed to the pressure connection. Under internal pressure the cross-section tends to become circular , so the tube straightens (uncoils) - free end moves .

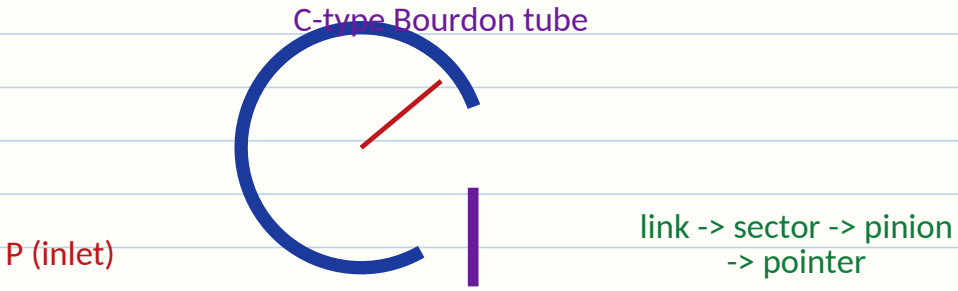


Fig 5.10: Bourdon pressure gauge

- Movement of the free end -> link -> quadrant (sector) -> pinion -> pointer (amplification of 1000 or more); hair spring removes backlash.
- Materials : phosphor bronze, brass, beryllium copper (low pressure); alloy steel / stainless steel (high pressure up to 7000 bar).
- Types : C-type (0 - 700 bar), spiral (more sensitive/ low pressure), helical (larger tip travel -> recorder, up to 7000 bar), twisted (for very high pressure / electrical pick-up).
- Advantages : simple, cheap, rugged, wide range (1 - 7000 bar) ; Disadvantages : hysteresis , not for fast-changing pressure , vibration damages; temperature effect; low-pressure limit ~ 0.5 bar (use diaphragm).

7.2 Diaphragm

A thin flexible disc (flat, corrugated, or dished) clamped at the edge ; pressure difference across it causes deflection of the centre (linear for small deflection: $y = 3 (1 - \nu^2) P R^4 / (16 E t^3)$).

- Flat (metal , linear for small deflection), corrugated (larger linear deflection , use for low pressure), slack diaphragm (rubber/fabric + spring, very low pressure: draft gauges); sensing : strain gauges bonded on the diaphragm (full bridge), LVDT , capacitive (variable gap) , piezo-resistive silicon (MEMS).
- Advantages : small, fast response (high natural frequency) , good for low and medium pressures (0.01 - 1000 bar) , differential pressure cells (DP transmitters, flow measurement). Disadvantages : limited deflection , temperature , non-linearity for large deflection.

7.3 Bellows and capsule

- Bellows : corrugated metal tube (brass, phosphor bronze, stainless steel) that expands/contracts axially with pressure against a spring : deflection $x = P A / k$ (A = effective area, k = spring rate). Used for low pressure (0.01 - 10 bar) , differential pressure (two bellows), pressure switches , aircraft altimeters .

- **Capsule (aneroid): two corrugated diaphragms joined at the edges** (disc-shaped chamber); **large deflection, very sensitive** ; stacked capsules **sum** the motion - used in **barometers (aneroid)** , **altimeters** , **vacuum gauges** .

Element	Typical range	Use
Bourdon tube	1 - 7000 bar	general purpose gauges
Diaphragm	0.01 - 1000 bar	low/medium, transmitters, DP
Bellows	0.01 - 10 bar	low pressure, switches, DP
Capsule	mbar - 1 bar	barometers, altimeters, vacuum

Quick Revision / Important Questions

1. Direct force measurement: analytical and platform balance; load cell, cantilever, proving ring.
2. Torque: torsion bar, servo-controlled and absorption dynamometers (Prony, rope brake).
3. Pressure standards, dead-weight gauge; manometers (U-tube, inclined).
4. High pressure (Bridgman) and low pressure (McLeod, Pirani, ionization).
5. Elastic pressure transducers: Bourdon tube, diaphragm, bellows, capsule.